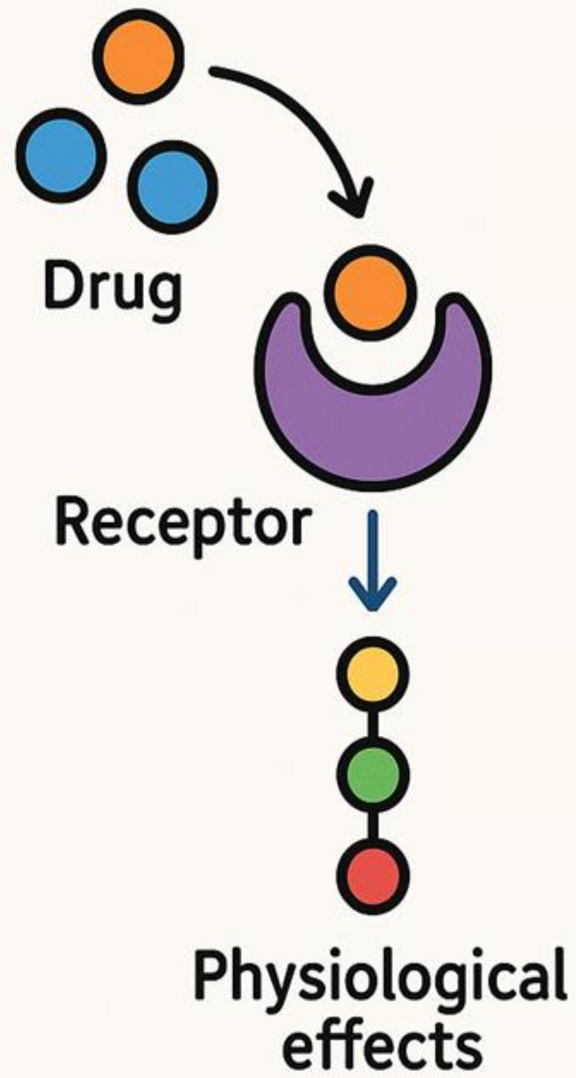
Basic Principles of Medicine-1 (BPM1)

FTM 30: Pharmacodynamics



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Fall Term 2025

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Objectives

SOM.MK.1.BPM1.1.FTM.4.PHAR.0001	Define the following terms: drug, pharmacology, pharmacodynamics, pharmacokinetics, pharmacotherapy and toxicology.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0002	Define drug receptor and list the different classes of receptors with which drugs interact.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0003	Give examples of drugs whose actions are not mediated by binding to receptors.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0004	Describe and interpret graded dose-response curves. Define E_{max} and EC_{50}.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0005	Describe and interpret drug-receptor binding curves. Define K_D and B_{max}. Define affinity.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0006	Define spare receptors and signal amplification.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0007	Compare and contrast *in vitro* experiments with whole animal experiments.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0008	Define efficacy and potency. Indicate the positions on a graded dose-response curve that are used to define drug potency and efficacy.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0009	Define agonist, antagonist, full agonist, partial agonist, and inverse agonist.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0010	Describe the different mechanisms of receptor and nonreceptor antagonism.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0011	Describe and interpret quantal dose-response curves. Define ED_{50}, TD_{50} and LD_{50}.
SOM.MK.1.BPM1.1.FTM.4.PHAR.0012	Explain therapeutic index and therapeutic window.

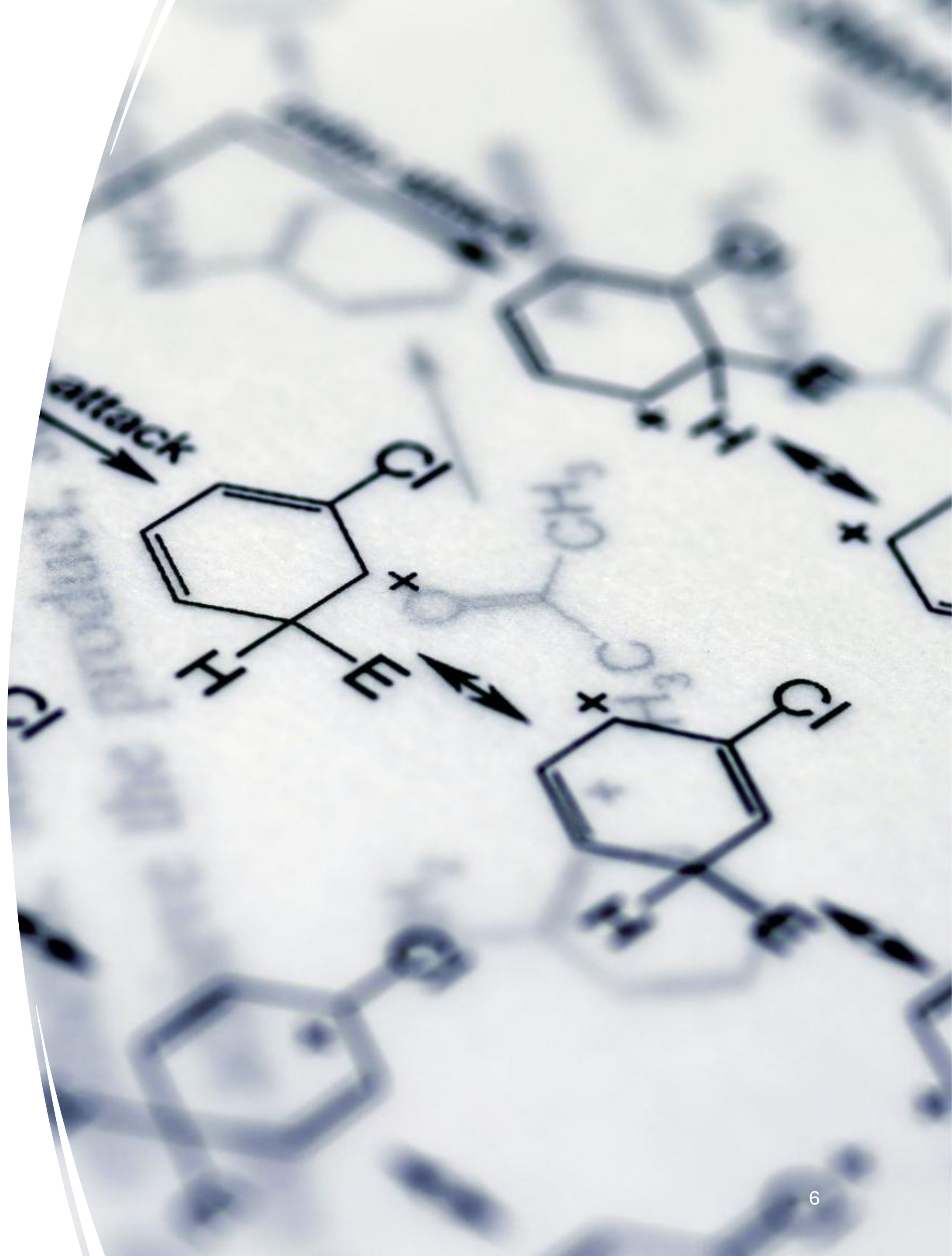
A Few Definitions

Drug:

any substance that when administered to a living organism, produces a biological effect.

A Few Definitions

Pharmacology: the study of how the function of living systems is affected by chemical agents.



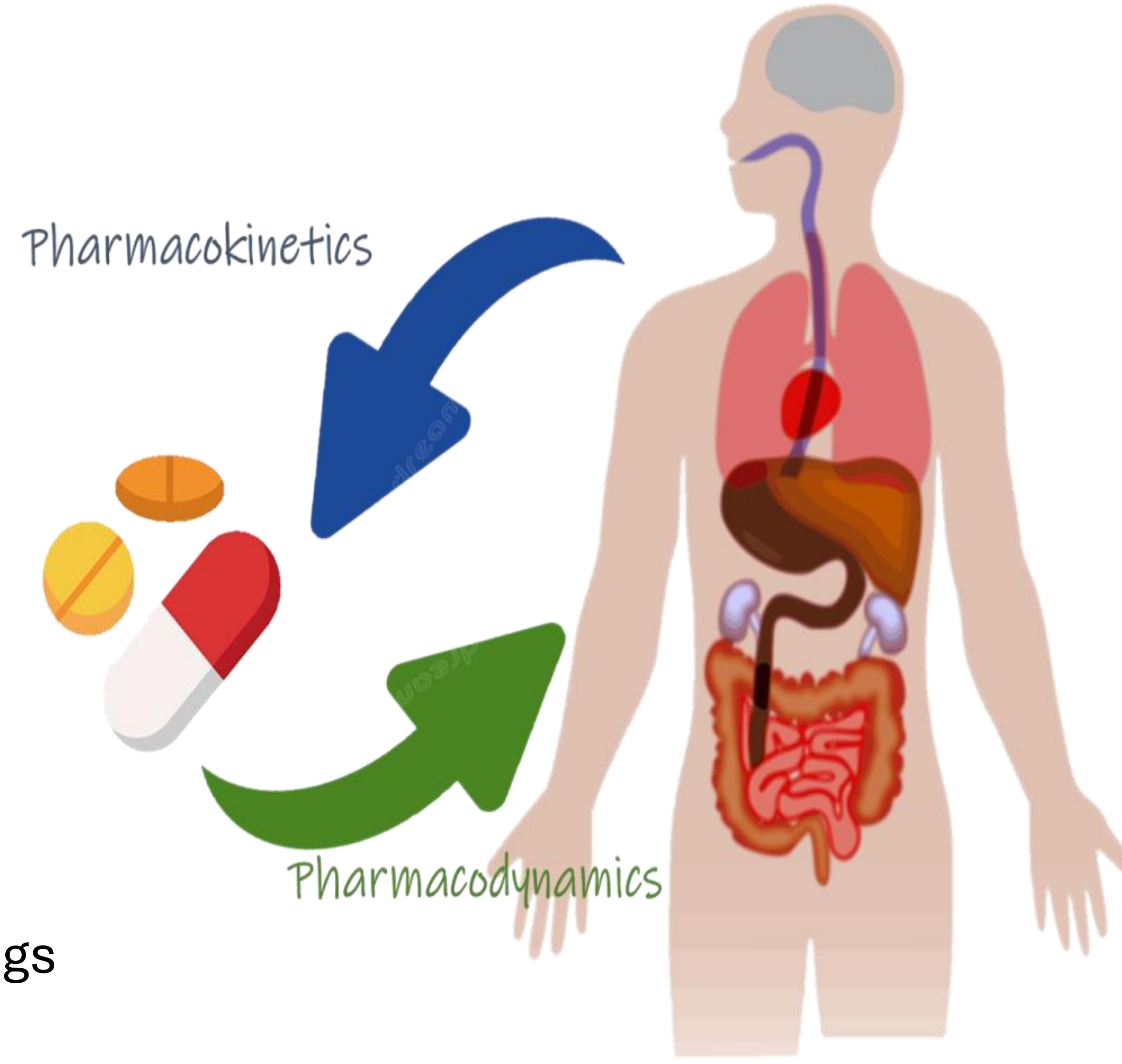
A Few Definitions

Pharmacokinetics:

the study of absorption, distribution, metabolism and excretion of drugs.

Pharmacodynamics:

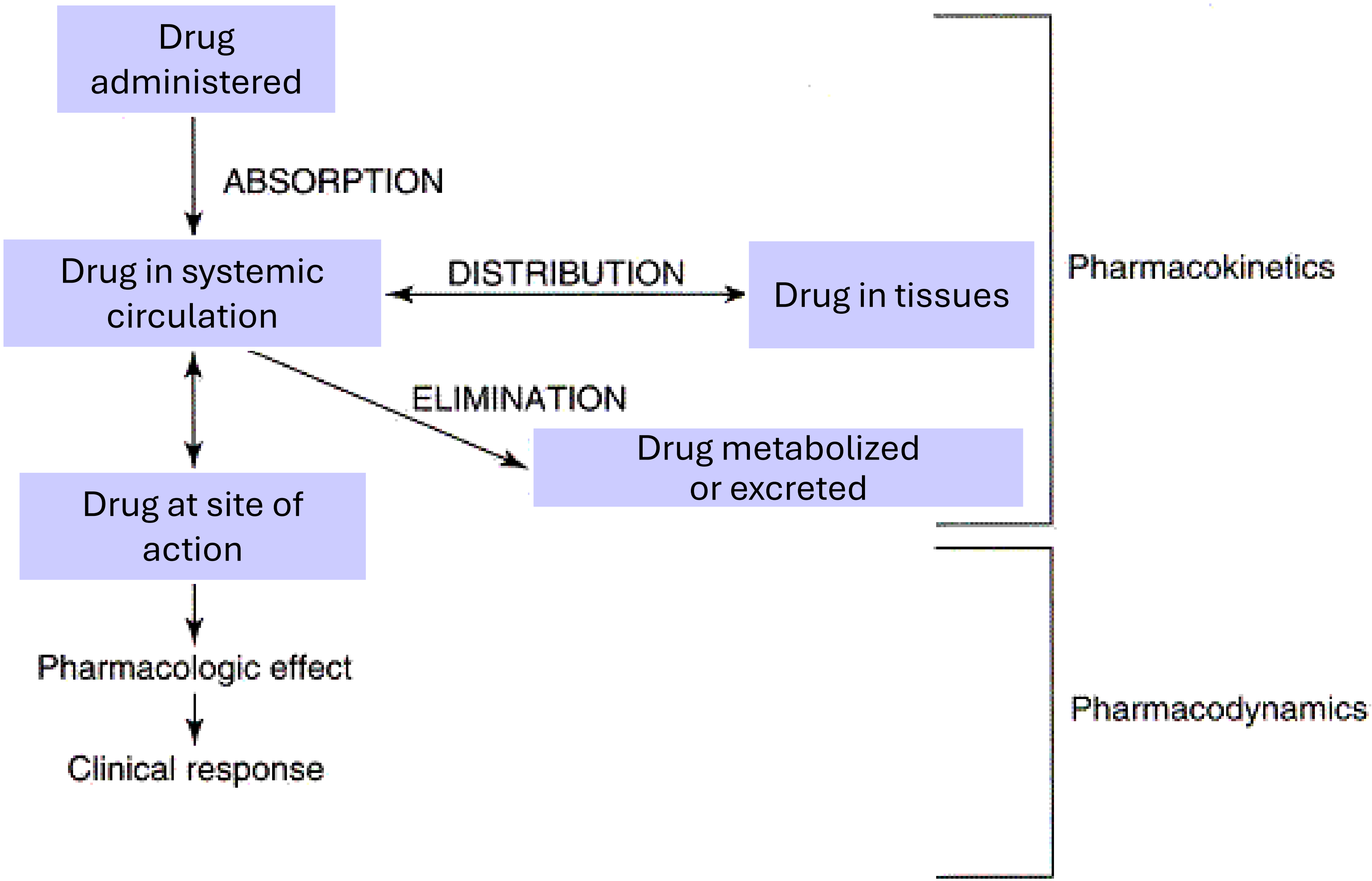
the study of effects of drugs and their mechanisms of action.



A Few Definitions

- ***Clinical pharmacology* or *Pharmacotherapeutics:*** the study of the use of drugs in the prevention and treatment of disease.
- ***Pharmacotherapy:*** use of drugs to treat disease.
- ***Toxicology:*** the study of adverse effects of drugs.

Pharmacokinetics & Pharmacodynamics



Drug Actions

- Drugs are chemicals that alter basic processes in body cells.

- They can stimulate or inhibit normal cellular functions.

Mechanisms of Drug Action

Drug molecules must **bind** to macromolecular components of the organism in order to produce an effect.

John Newport Langley (1852–1925)

Mechanisms of Drug Action

- The component of the organism with which a drug interacts is called a ***drug receptor*** **or** ***drug target***.
- **Most drug receptors are proteins.**

Paul Ehrlich (1854–1915)

Mechanisms of Drug Action

- **The majority of drug receptors are physiological receptors**, which mediate the actions of neurotransmitters and hormones.

- Other classes of proteins, such as enzymes, transport proteins, and structural proteins are also drug receptors.

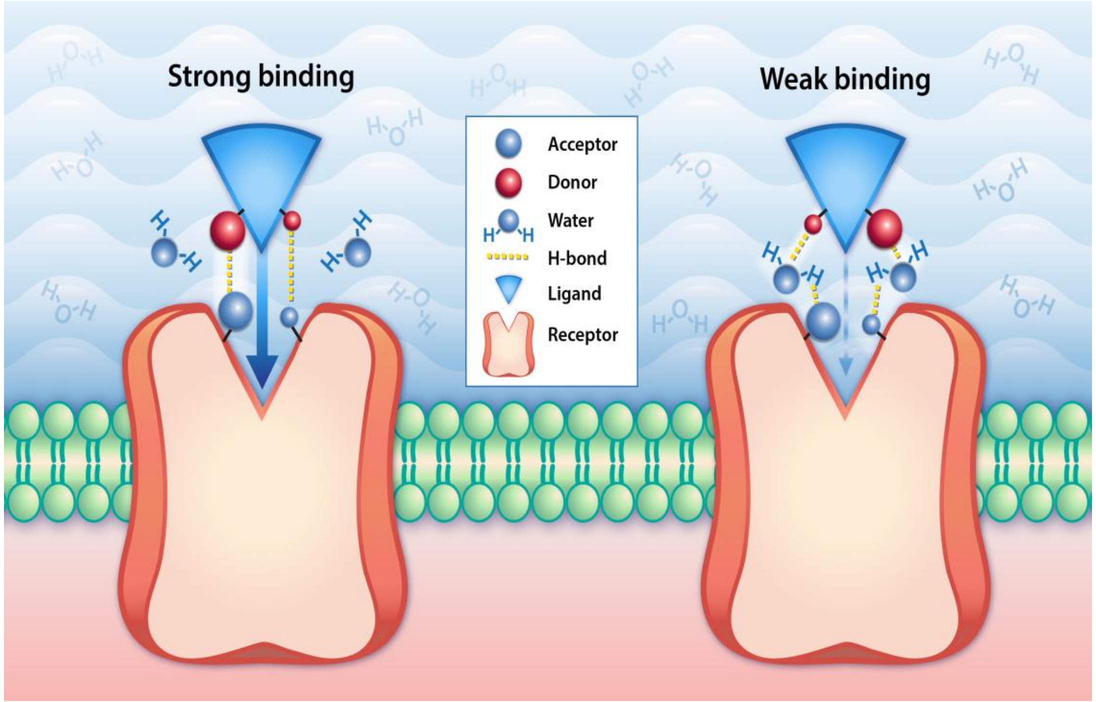
Mechanisms of Drug Action

- **To initiate a cellular response, a drug must first bind to a drug receptor.**

- In most cases, drugs bind to their receptor by forming hydrogen, ionic, or hydrophobic bonds with the receptor.

- These weak bonds are reversible.

- In a few cases, drugs form covalent bonds with their receptor.



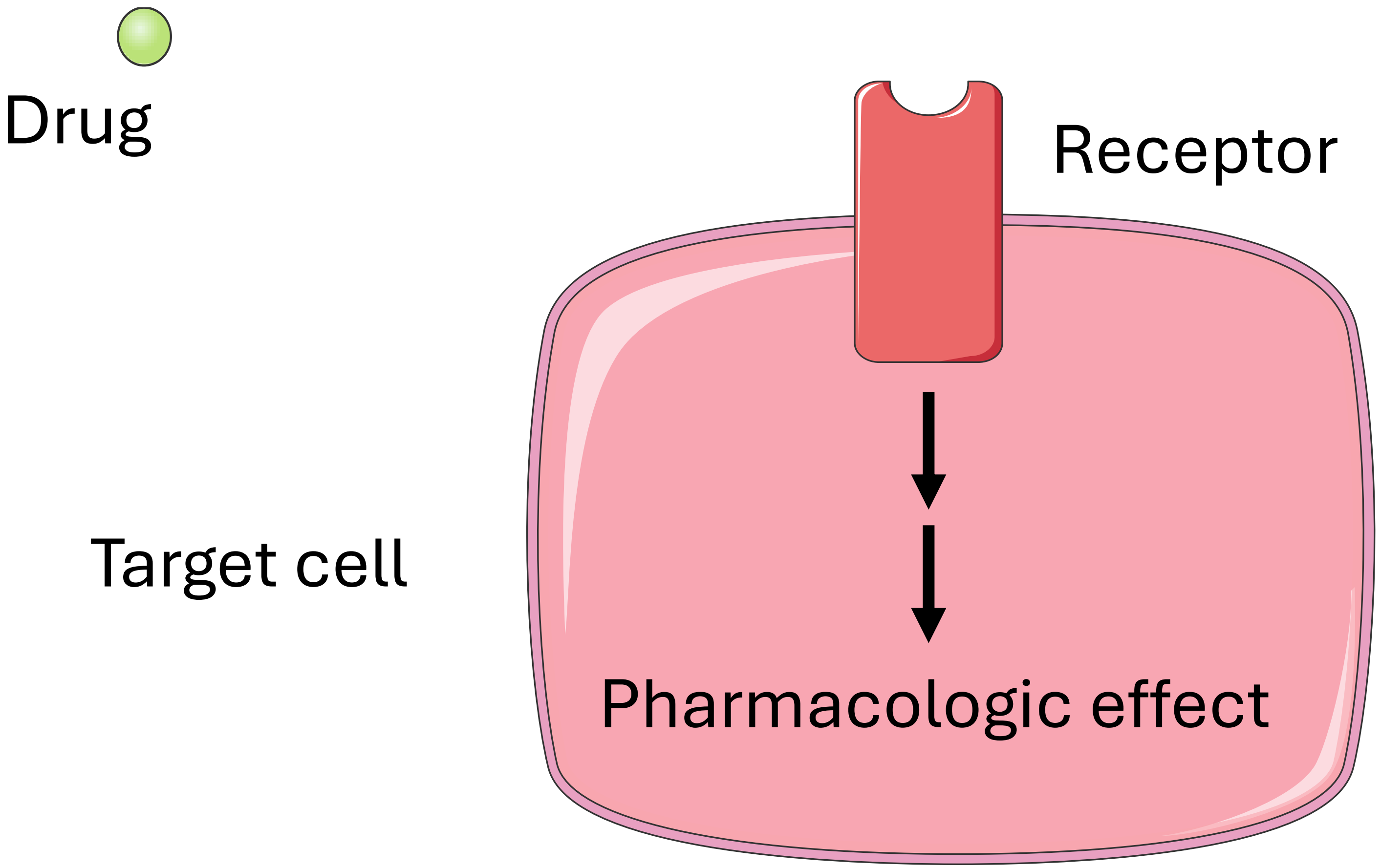
Mechanisms of Drug Action

- **Drug receptors are responsible for the selectivity of drug action.**

- The size, shape, and charge of a drug determine whether it will bind to a particular receptor or set of receptors.

- For a drug to be useful as a therapeutic tool it must act **selectively** on particular types of receptors, ie, it must show a high degree of **binding selectivity.**

Mechanisms of Drug Action



Major Types of Drug Targets

- G-protein-linked receptors
- Enzyme-linked receptors
- Ion Channels
- Nuclear receptors
- Enzymes
- Transporters
- Structural proteins

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G Protein-linked Receptors

- Approximately 60 % of prescription drugs act by binding to G protein linked receptors.
- Examples:
    - **Albuterol**, a β_2-agonist, is used for asthma.
    - **Propranolol**, a β-antagonist, is used for hypertension.

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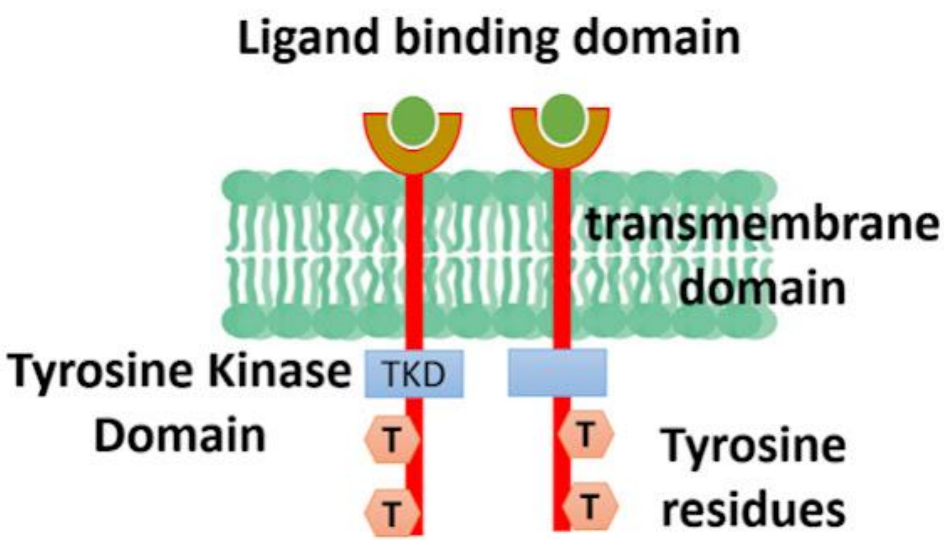
Enzyme-Linked Receptors

- The largest group is the **receptor tyrosine kinase family.**
- This class of receptors includes the
  - **Insulin receptor**
  - **Epide**



Tyrosine Kinase Receptors

- **Tyrosine kinase receptors play an important role in cellular growth and differentiation.**
- Gain-of-function mutations in these receptors can lead to cancer.
- **Inhibitors of tyrosine kinase receptors are used as anticancer agents.**
- An example is **imatinib**, which is effective for leukemia.

Major Types of Drug Targets

- G-protein-linked receptors
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Ion Channels

- Several classes of drugs act by altering the conductance of ion channels.
- Examples are:
  - **Local anesthetics**
  - **Sedative-hypnotics**
  - **Antiepileptics**

Major Types of Drug Targets

- G-protein-linked receptors
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Nuclear Receptors

- **Nuclear receptors are intracellular.**
- Nuclear receptors regulate the expression of genes controlling metabolism and development.
- Members of the nuclear receptor superfamily include receptors for:
  - **Steroid hormones**
  - **Thyroid hormone**
  - **Vitamin D**

Major Types of Drug Targets

- G-protein-linked receptors
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Enzymes

- Enzymes are a common drug target.
- Most drugs that target enzymes act by **inhibiting enzyme activity**.
- Examples:
  - **Aspirin**
  - **Ibuprofen**
  - **Omeprazole**

Major Types of Drug Targets

- G-protein-linked receptors
- Enzyme-linked receptors
- Ion Channels
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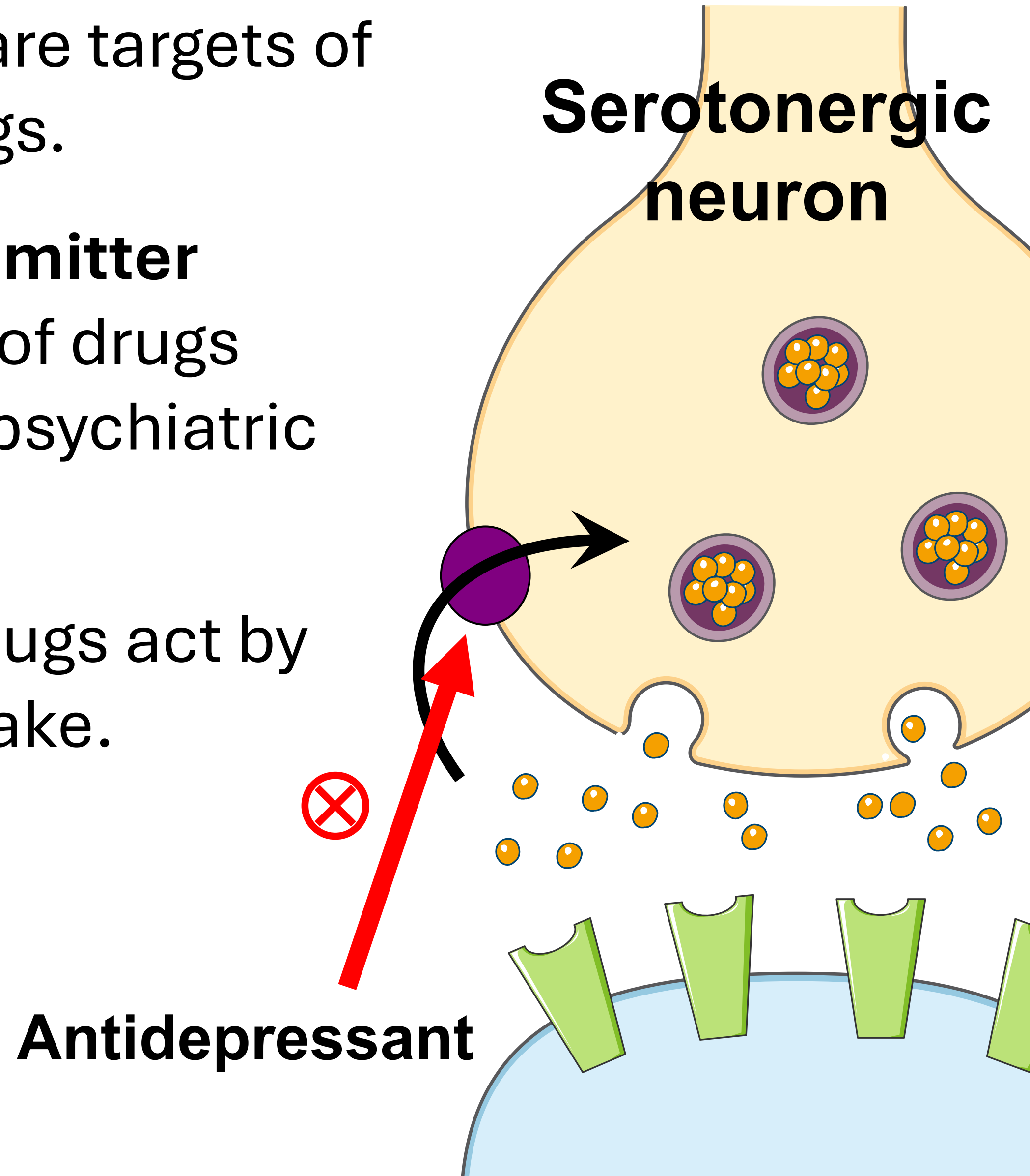
Transporters

- Membrane transporters are targets of many clinically used drugs.
- For example, **neurotransmitter transporters** are targets of drugs used in the treatment of psychiatric disorders.
- Some **antidepressant** drugs act by blocking serotonin reuptake.



Major Types of Drug Targets

- G-protein-linked receptors
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- Structural proteins

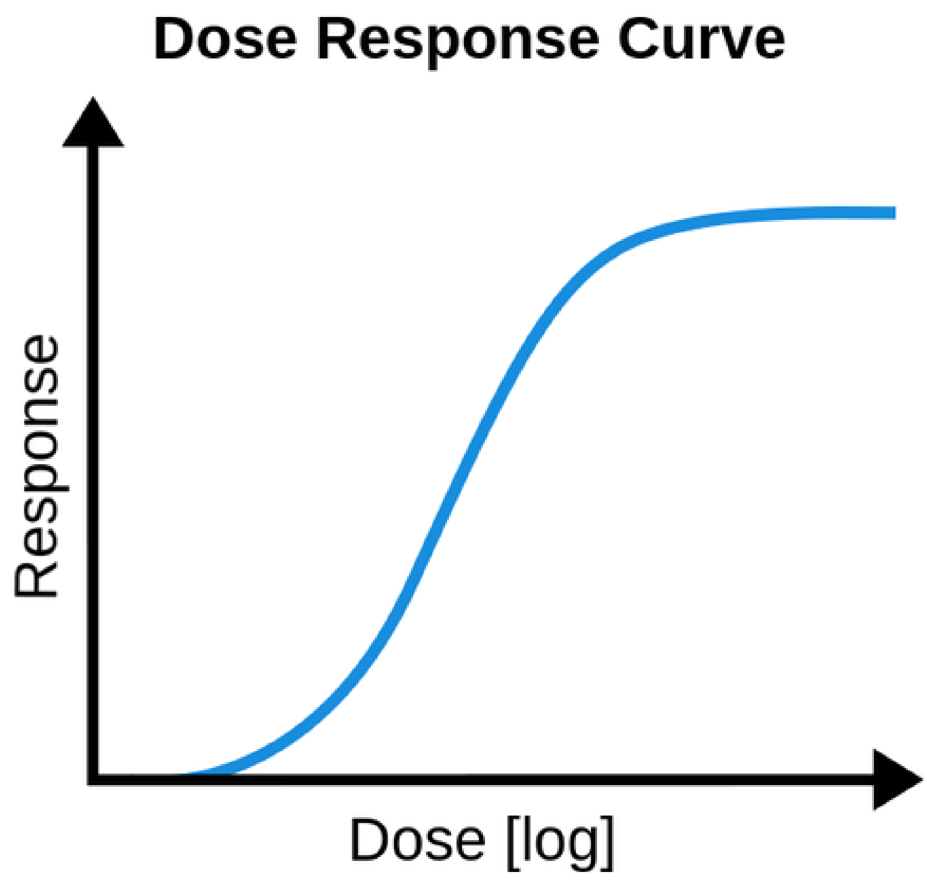
Structural Proteins

- Some **anticancer** drugs bind to **tubulin** and prevent the polymerization of this molecule into microtubules.
- As a consequence, cells are arrested in metaphase.
- An example is **vinblastine.**

Actions of Drugs Not Mediated by Binding to Receptors

- Some drugs do not act by interacting with macromolecular components of the organism.
- **Antacids** neutralize gastric acid.

DOSE-RESPONSE CURVES



Dose-Response Curves

- There are two major types of dose-response relationships: **graded and quantal.**

Graded Dose-Response Curves

- The relation between drug concentration and effect is described by a **hyperbolic** curve according to the following equation:

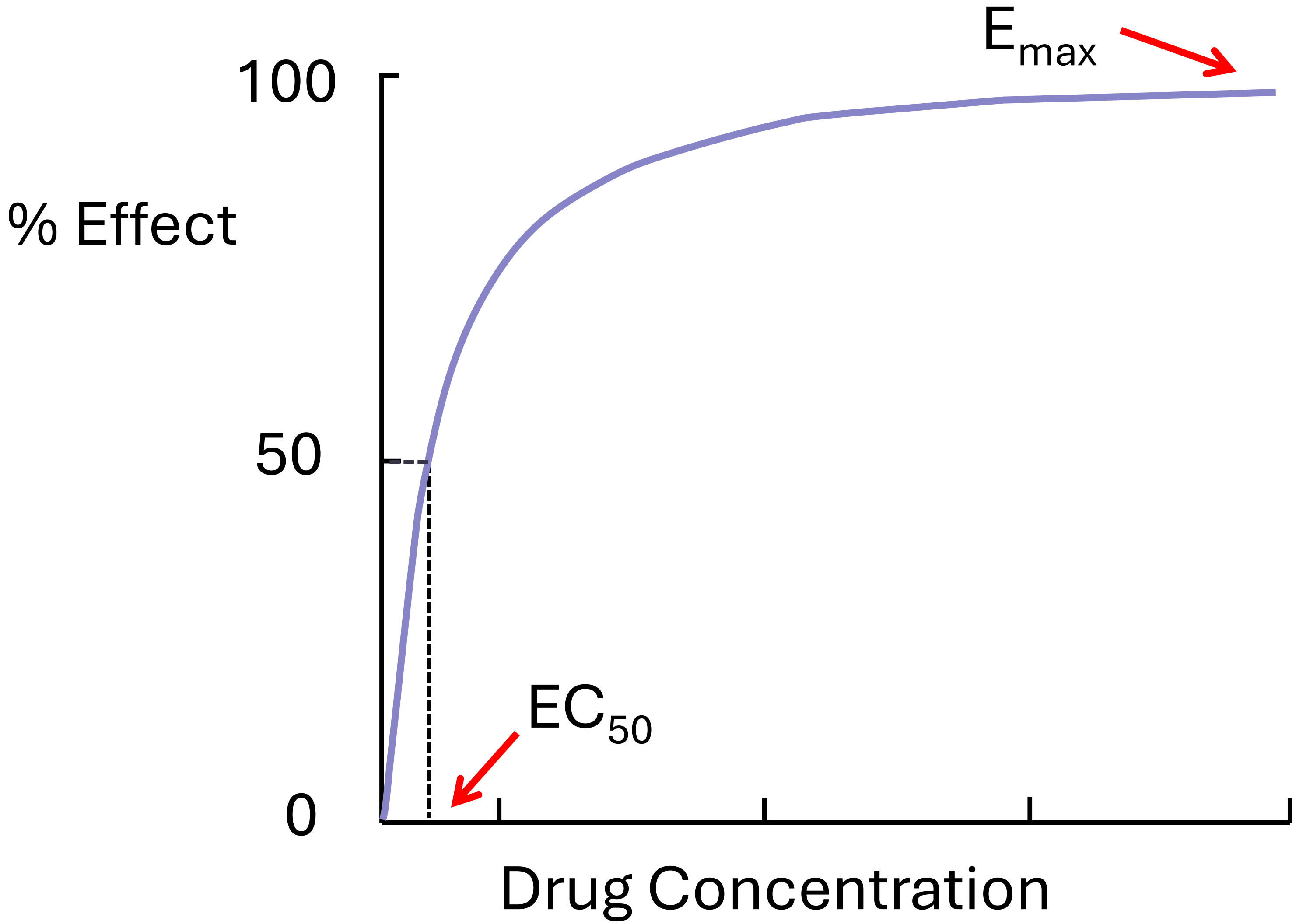
$$\mathbf{E = \frac{E_{max} \times C}{C + EC_{50}}}$$

E is the effect

E_{max} is the maximal response produced by the drug

EC_{50} is the drug concentration that produces 50% of the maximal effect.

Graded Dose-Response Curves



Receptor Binding Curve

- The relation between drug bound (B) to receptors and the concentration (C) of free drug is described by the equation:

$$B = \frac{B_{max} \times C}{C + K_D}$$

B_{max} is the total concentration of receptor sites.

K_D is the equilibrium dissociation constant.

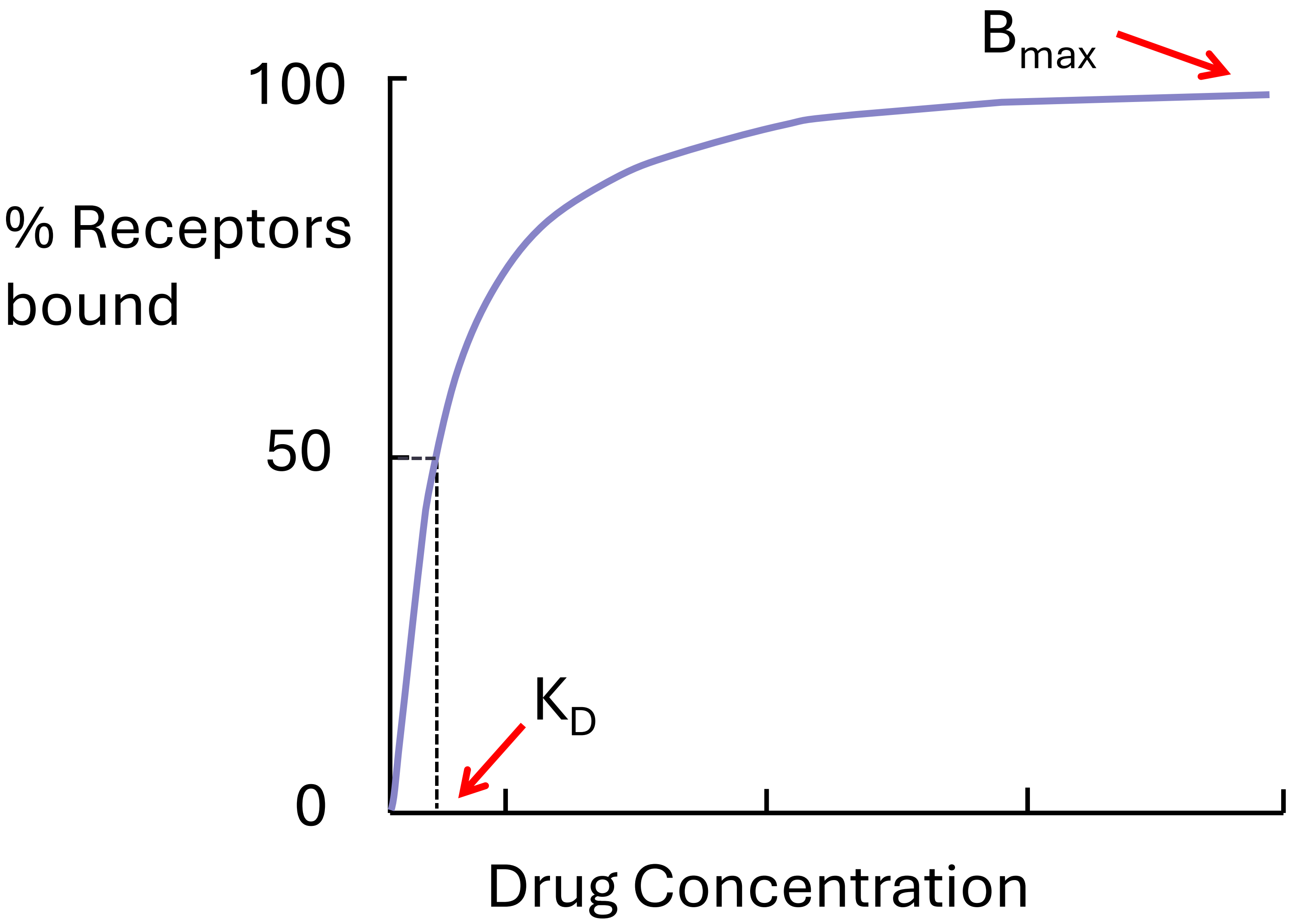
K_D is the concentration of drug required to occupy half of the receptors.

Receptor Binding of Drugs

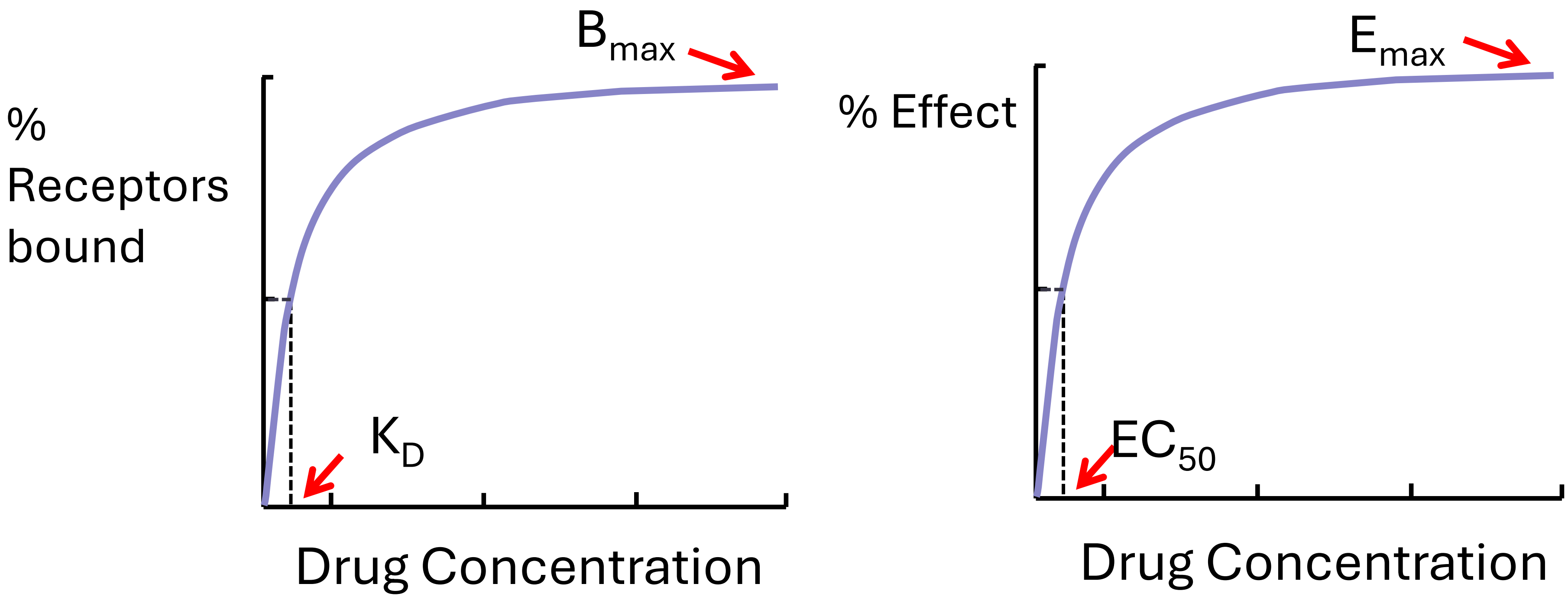
$$B = \frac{B_{max} \times C}{C + K_D}$$

- K_D characterizes the receptor's **affinity** for the drug**.**
- **Affinity** is the tendency of a drug to combine with its receptor: it is a measure of the strength of the **drug-receptor complex.**
- If K_D is low, binding affinity is high, and vice versa.

Receptor Binding of Drugs



Binding Curve & Dose Response Curve

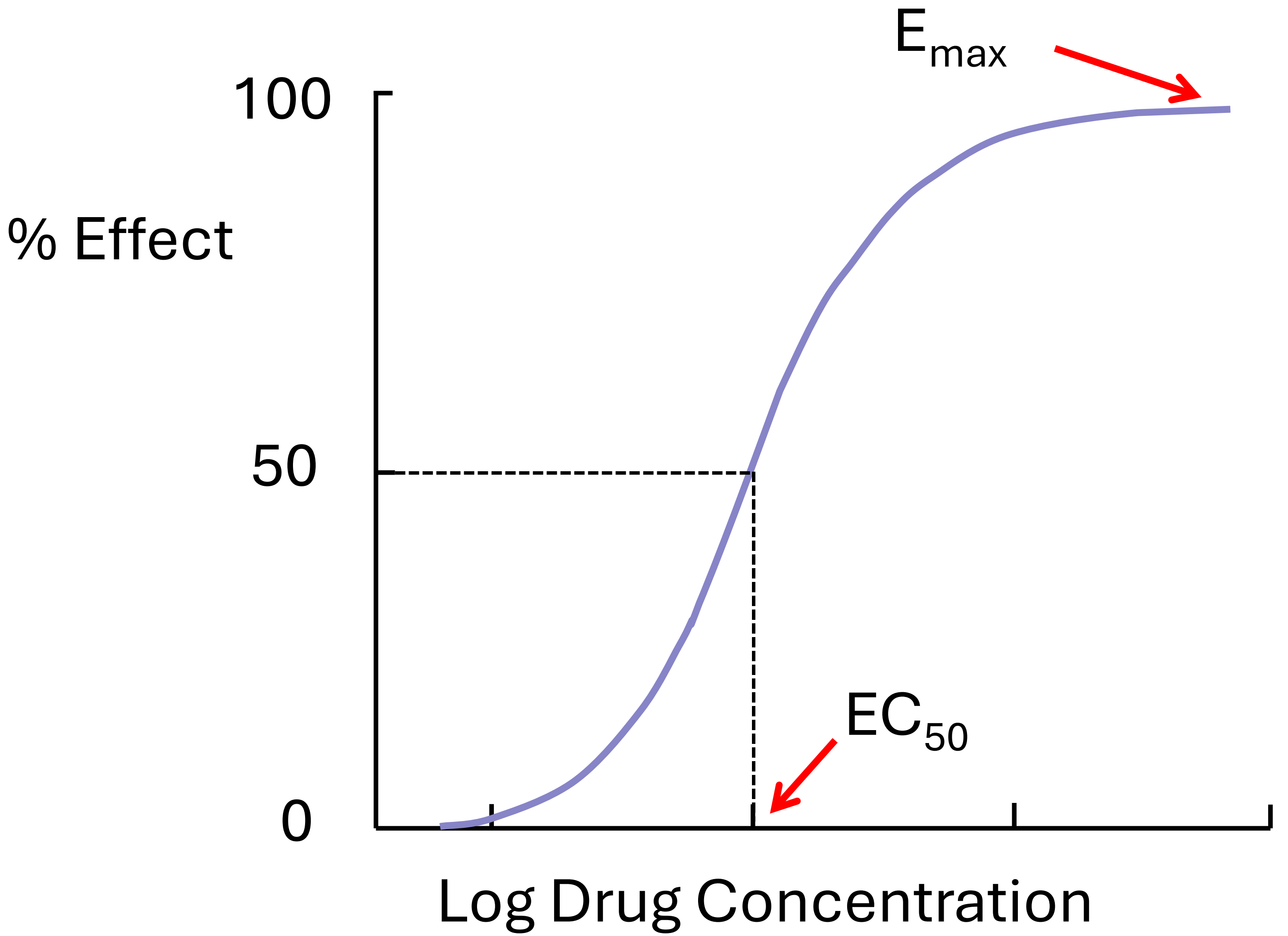


- The shape of the dose-response curve is similar to the binding curve because the response to the drug is a consequence of the binding of the drug to the receptor.

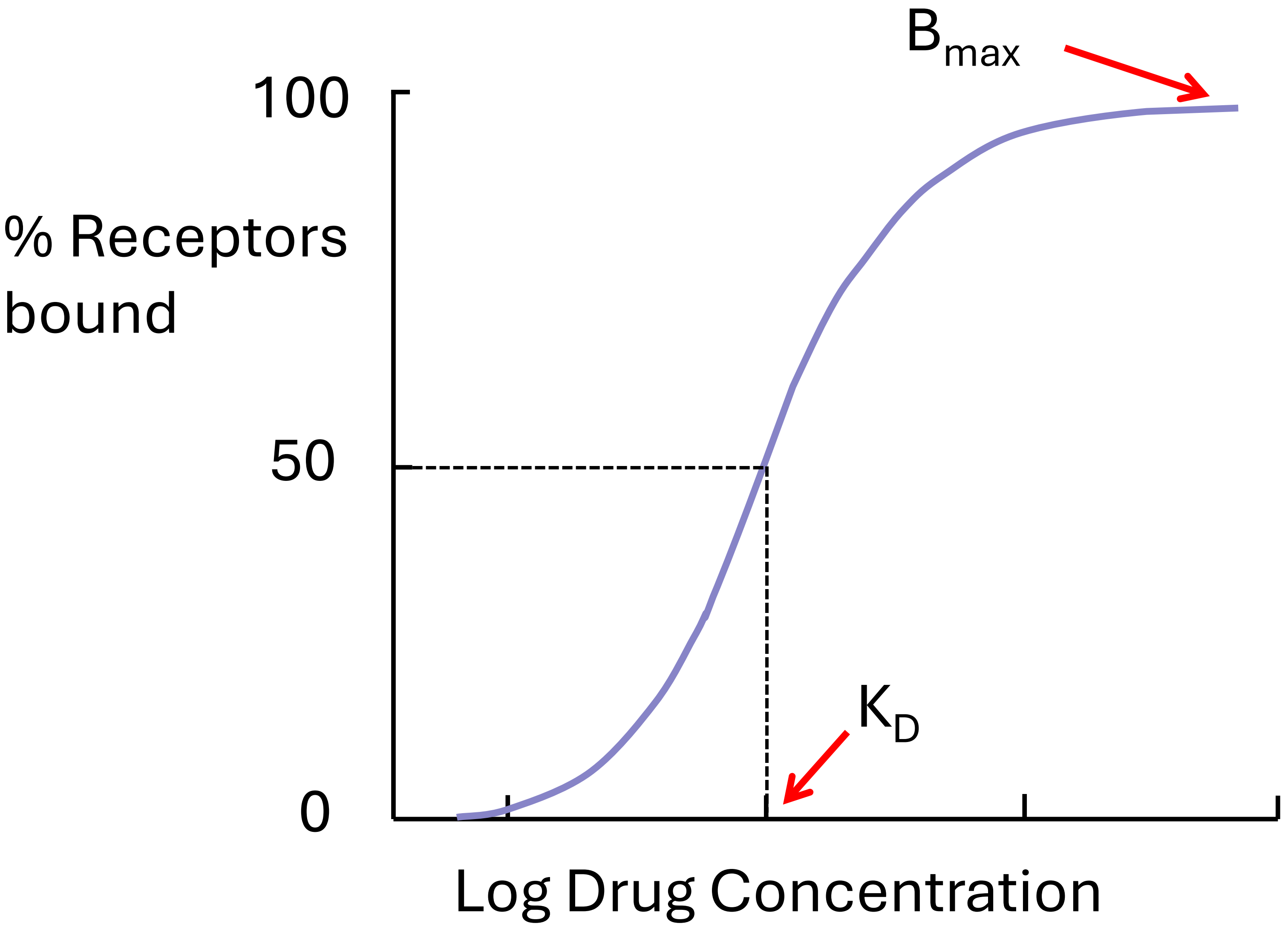
Graded Dose-Response Curves

- Dose-response data is frequently plotted as the drug effect against the **logarithm of the concentration.**
- This transforms the hyperbolic curve into a sigmoid curve.

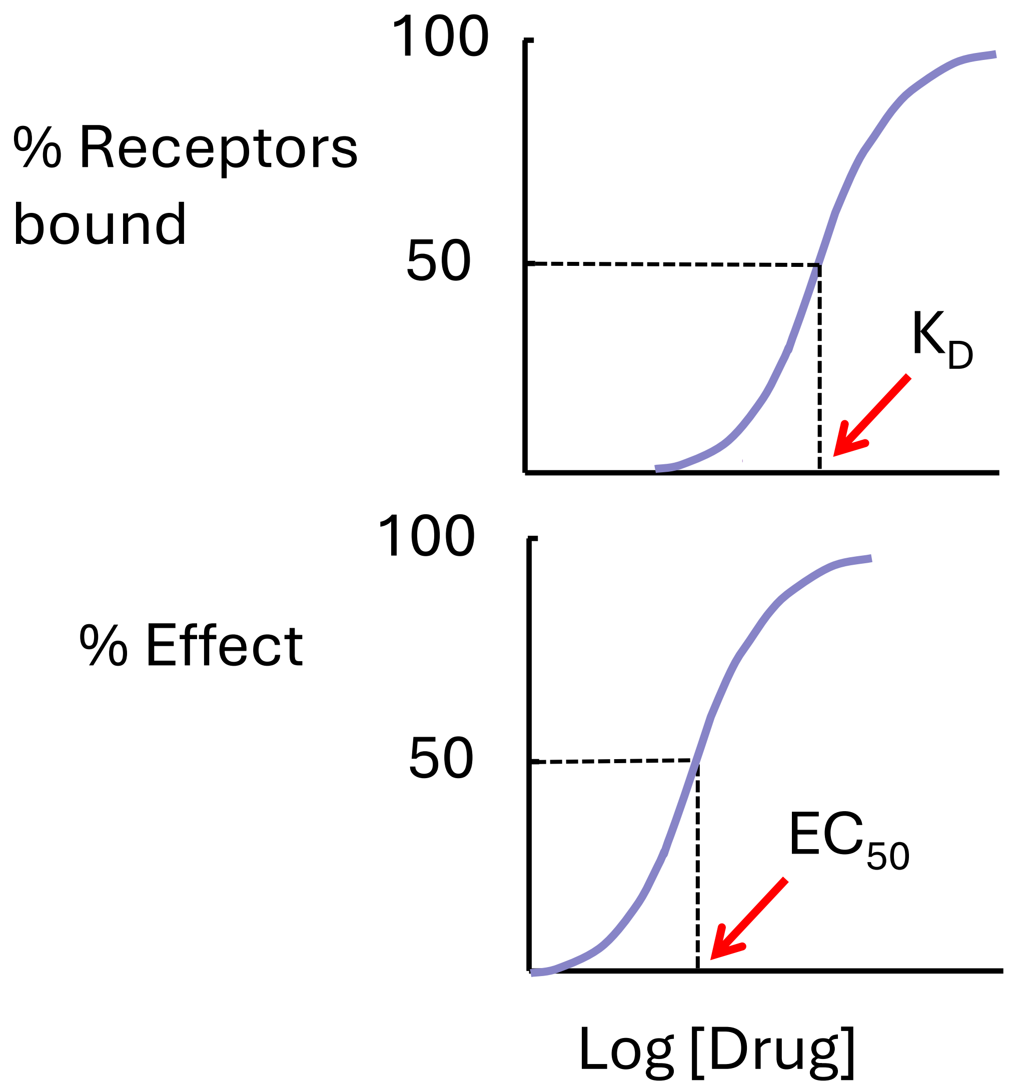
Graded Dose-Response Curves



Receptor Binding of Drugs



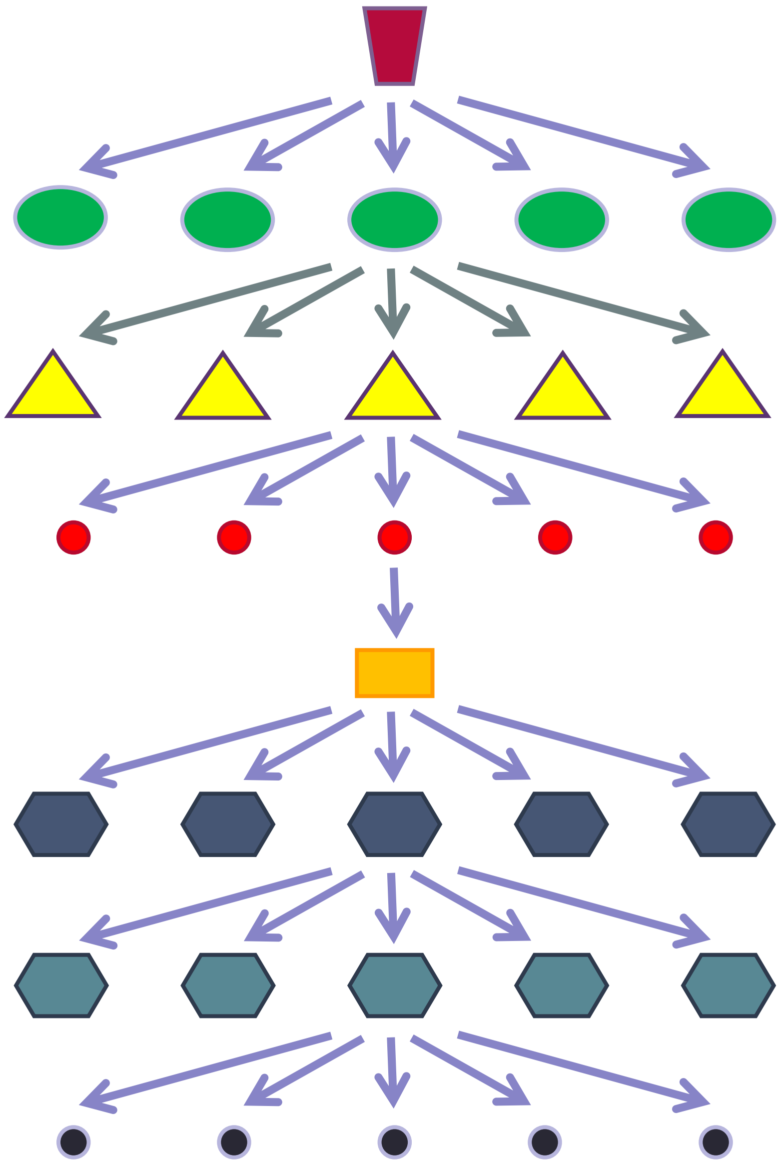
Spare Receptors & Signal Amplification



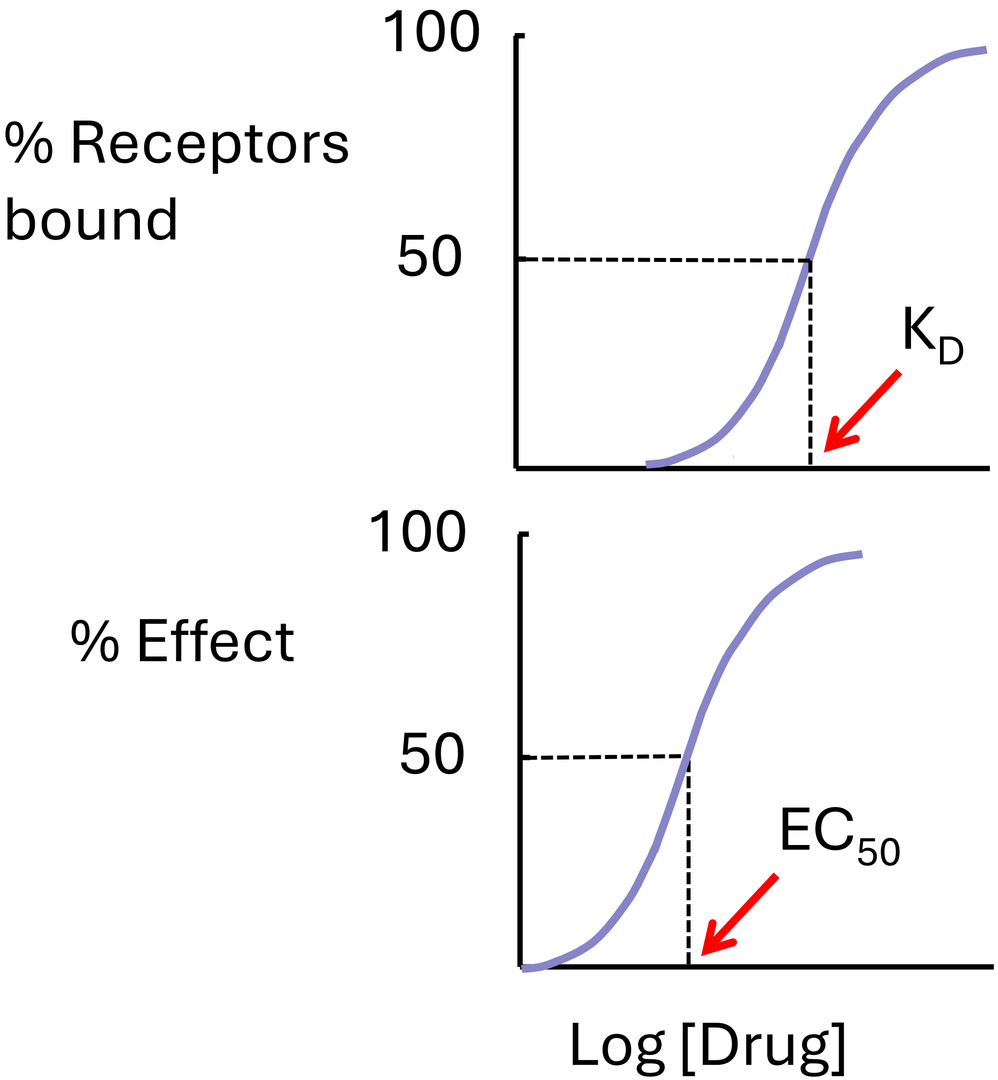
Spare Receptors & Signal Amplification

- **An agonist does not have to occupy all receptors to evoke a full response.**
- **Therefore EC_{50} is lower than K_D.**
- Because of this a certain number of receptors are said to be "spare."
- **The presence of spare receptors indicates that there is signal amplification.**

Spare Receptors & Signal Amplification



Spare Receptors & Signal Amplification

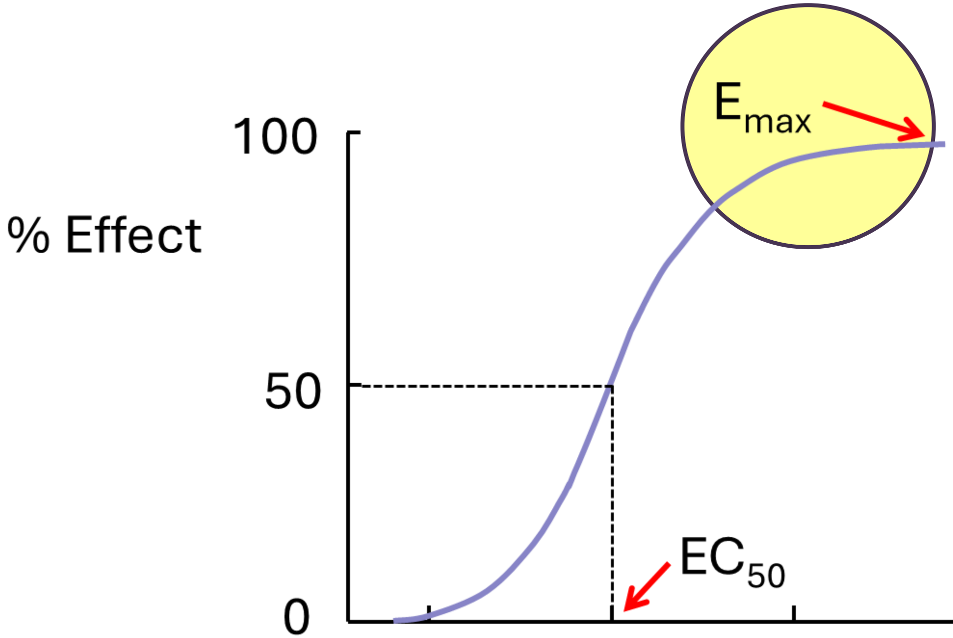


Efficacy & Potency

Efficacy

- **Efficacy is the magnitude of the response a drug produces.**

Maximal efficacy (sometimes referred to simply as efficacy), is the greatest effect a drug can produce (E_{max}).



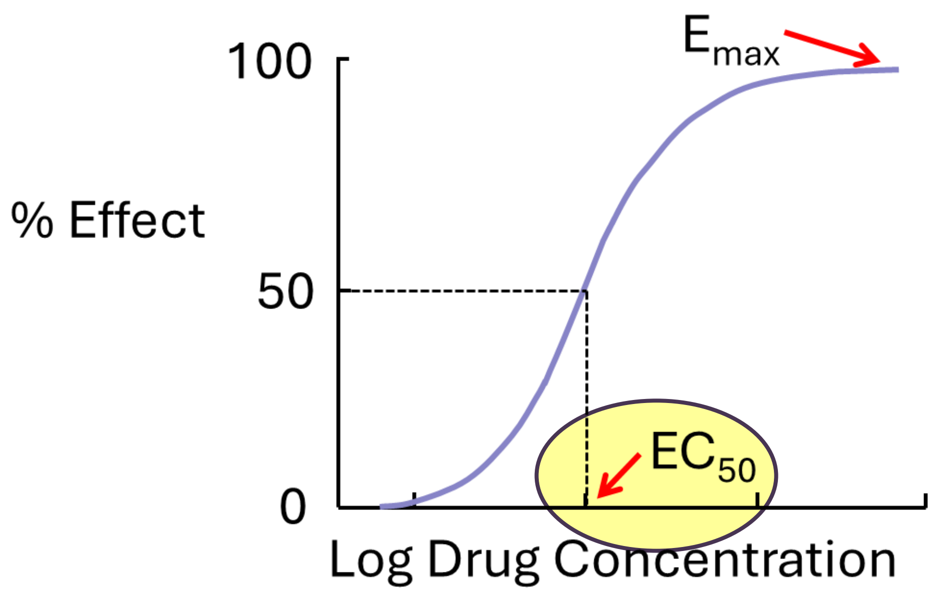
Efficacy & Potency

Potency

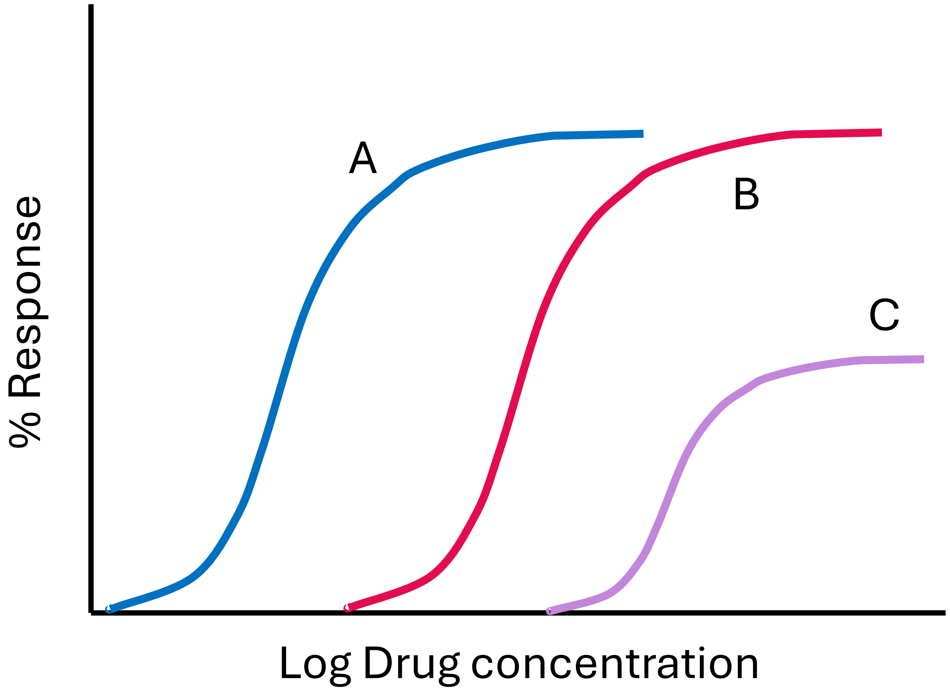
Potency is a measure of the concentration or amount of drug necessary to produce an effect of a given magnitude.

The EC_{50} of a drug is usually used to determine potency.

The potency of a drug depends both on the **affinity** and the **efficacy** of the drug.



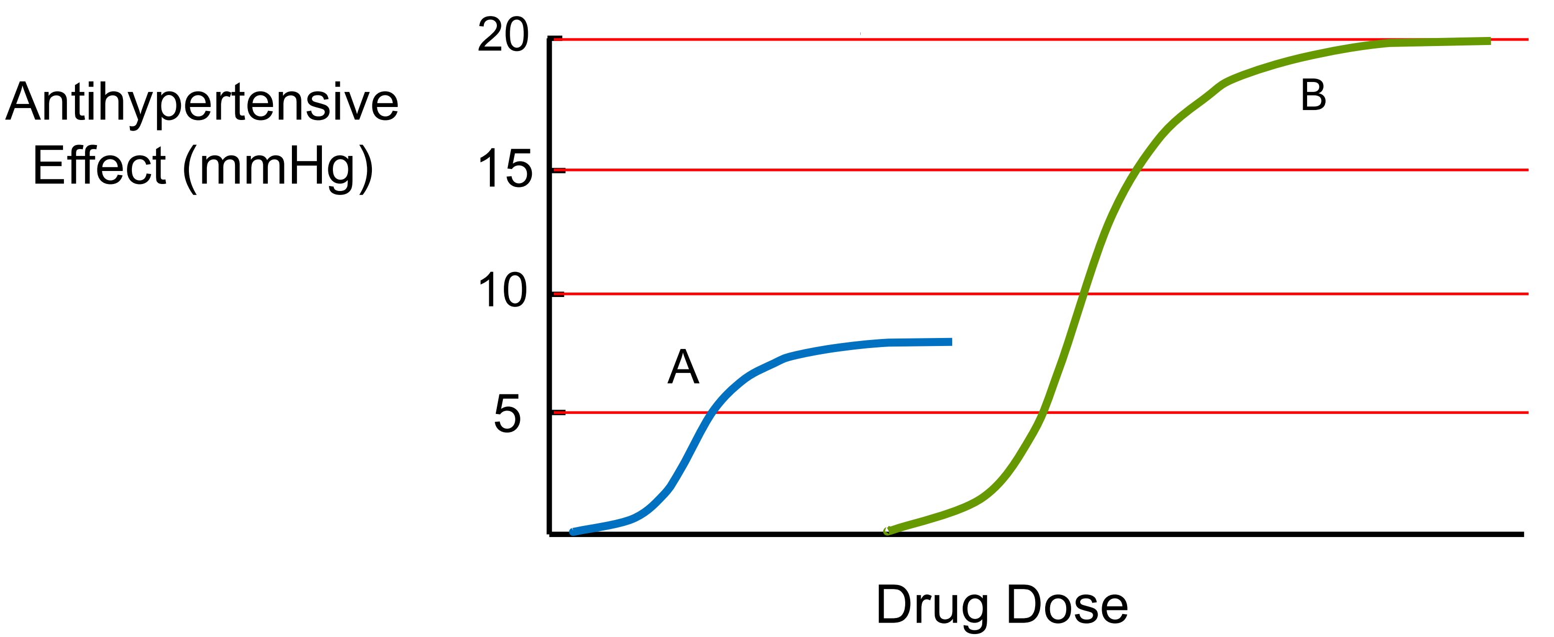
Efficacy & Potency



- Drug A is more potent than Drug B.
- Drug A and Drug B have equal efficacy.
- Drug C has lower potency and lower efficacy than Drugs A and B.

More about Efficacy & Potency

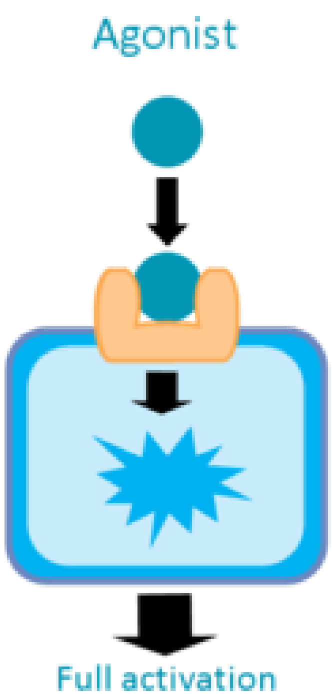
- A and B are antihypertensive drugs.
- At low responses A is more potent than B.
- At high responses B is more potent than A.
- Therefore, no comparisons can be made between A and B in terms of potency.

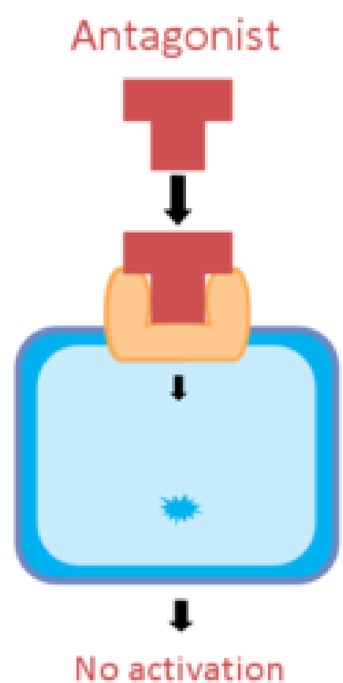


Agonists & Antagonists

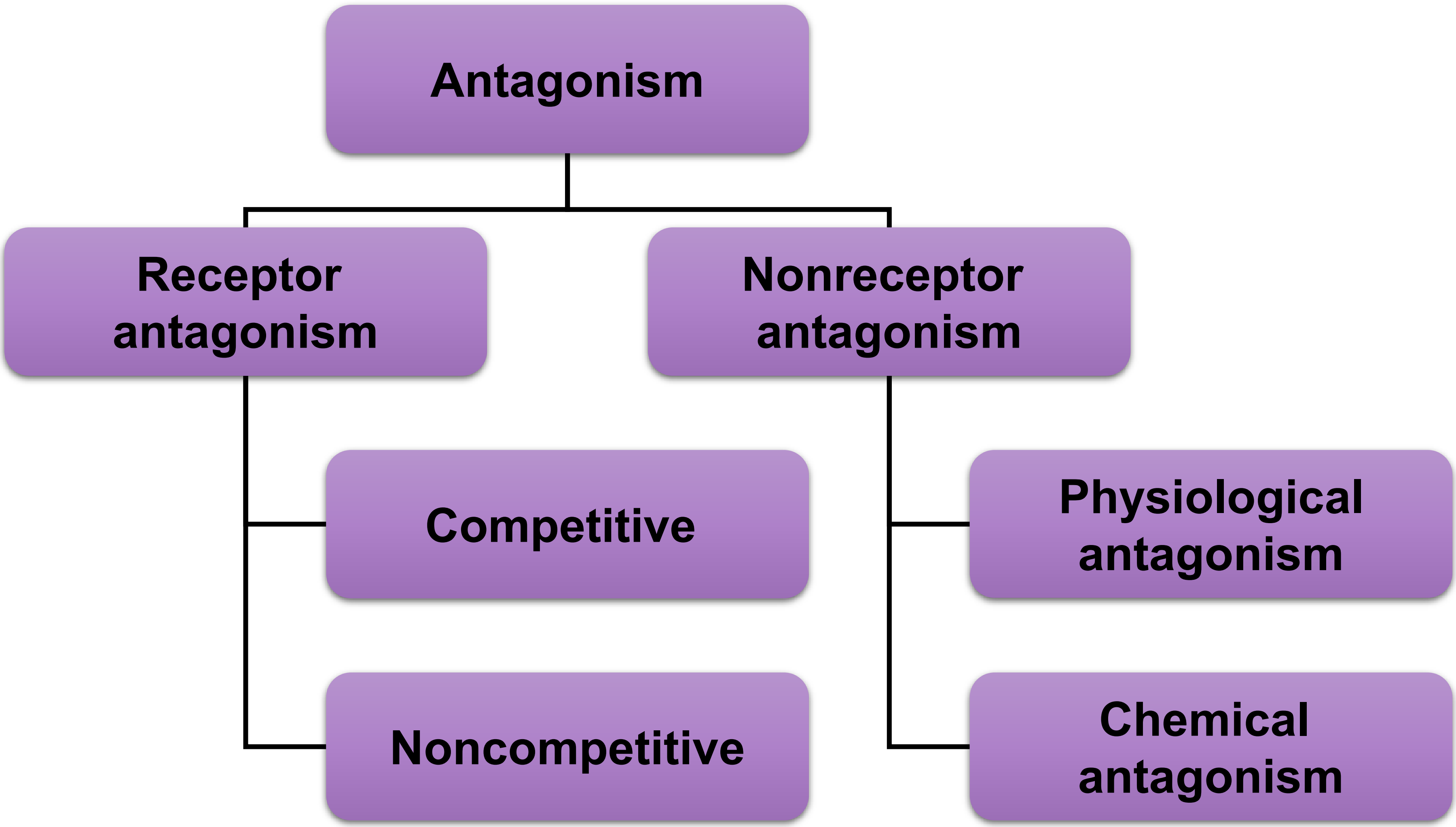
An agonist is a drug that binds to and activates a receptor in a way that brings about an effect.

An antagonist is a drug that blocks or reduces the action of an agonist.





MECHANISMS OF DRUG ANTAGONISM

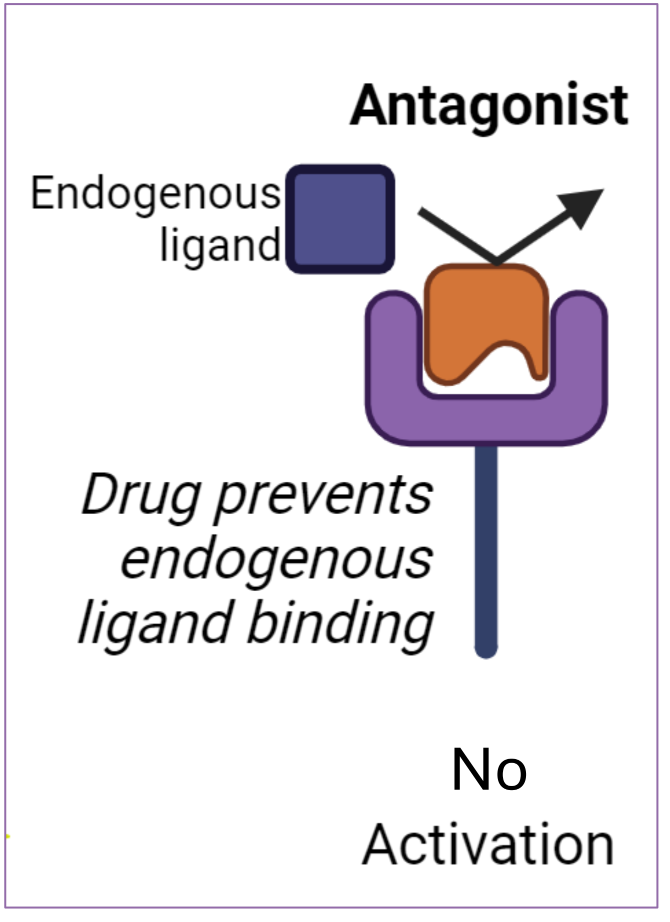


RECEPTOR ANTAGONISM

- **A receptor antagonist binds to the same receptor to which the agonist binds.**
- Receptor antagonists bind to receptors but do not activate them.
- They inhibit the action of an agonist but have no effect in the absence of the agonist.
- Agonists have receptor affinity and efficacy.
- Receptor antagonists have receptor affinity but lack efficacy.

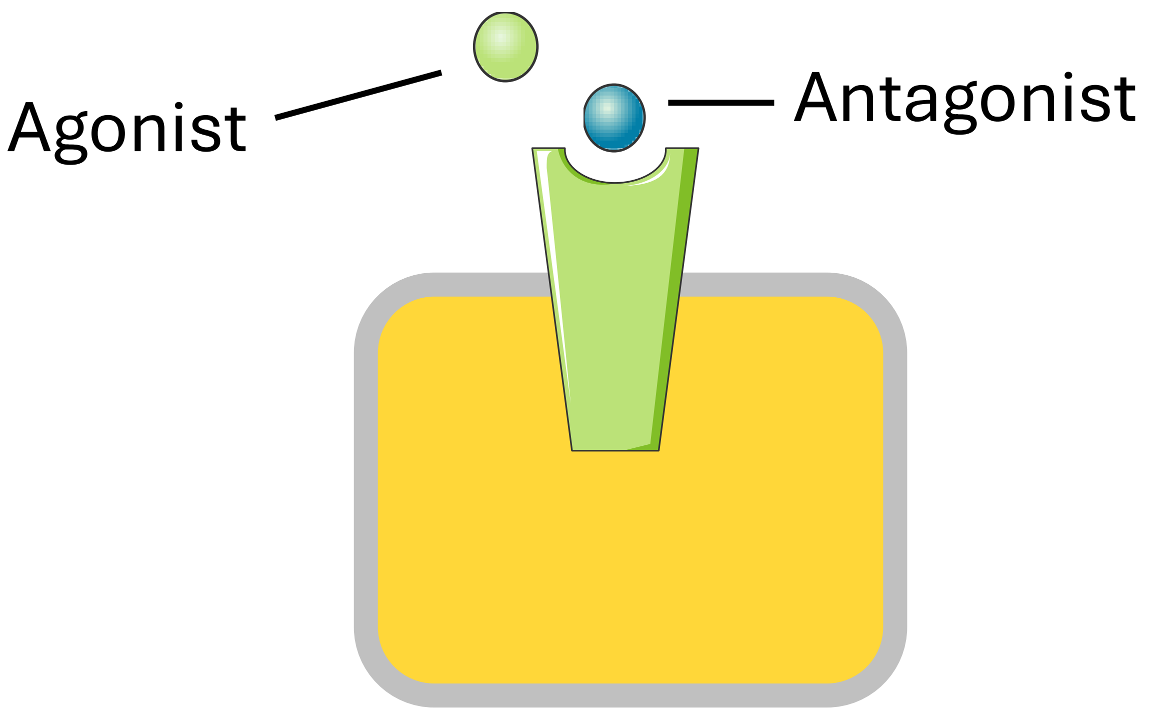
RECEPTOR ANTAGONISM

- Receptor antagonism can be competitive or noncompetitive.



Competitive Antagonism

- Competitive antagonists bind to the **agonist binding site** on the receptor.
- Binding of antagonist to the agonist binding site prevents the binding of agonist to the receptor.



Competitive Antagonism

- Binding of antagonist to the receptor may be **reversible or irreversible.**

Reversible Competitive Antagonism

- Reversible competitive antagonists are the most important class of antagonists.
- Many clinically used drugs belong to this class.

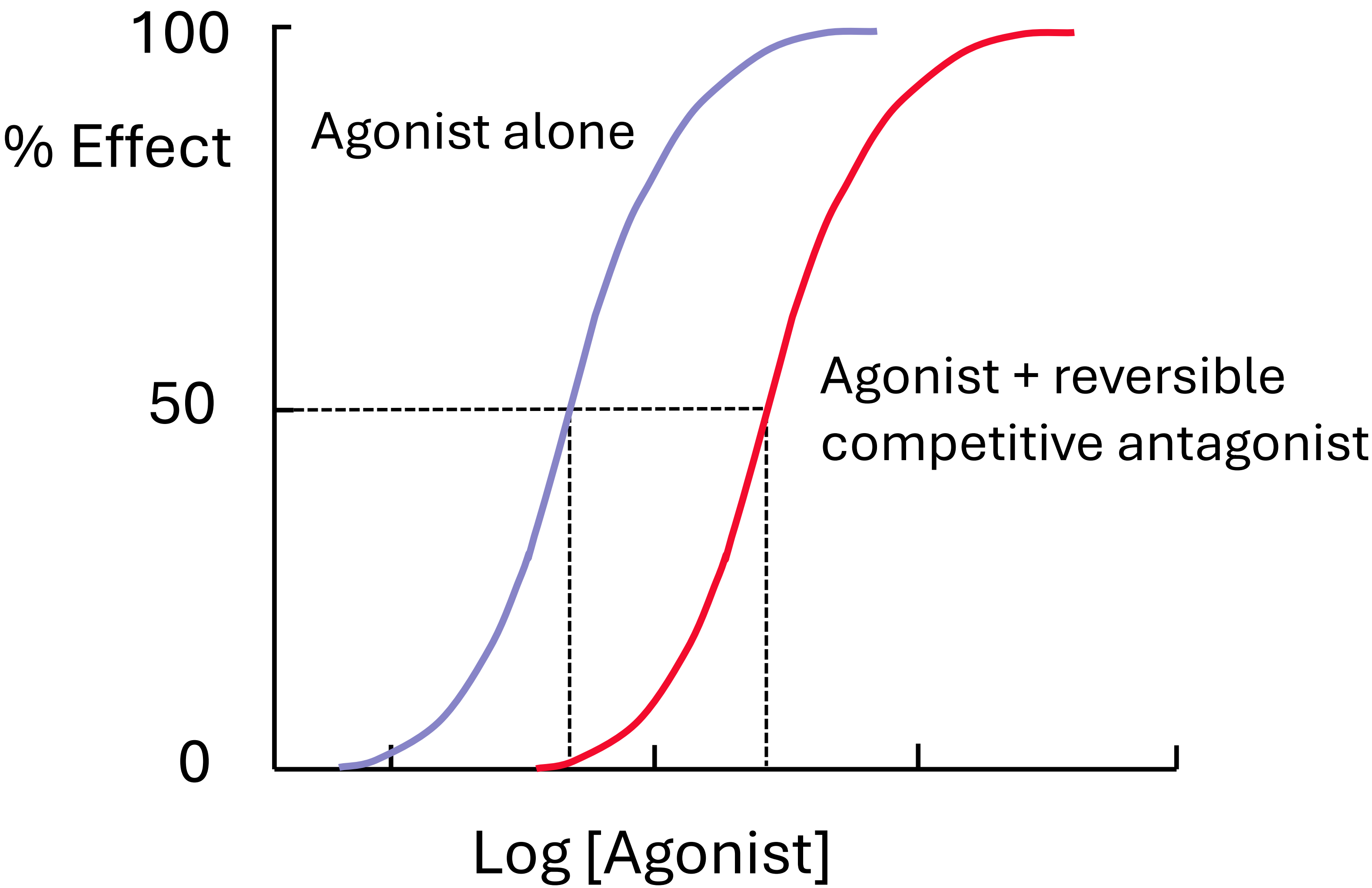
Reversible Competitive Antagonism

- The presence of the reversible competitive antagonist increases the concentration of agonist required for a given degree of response.

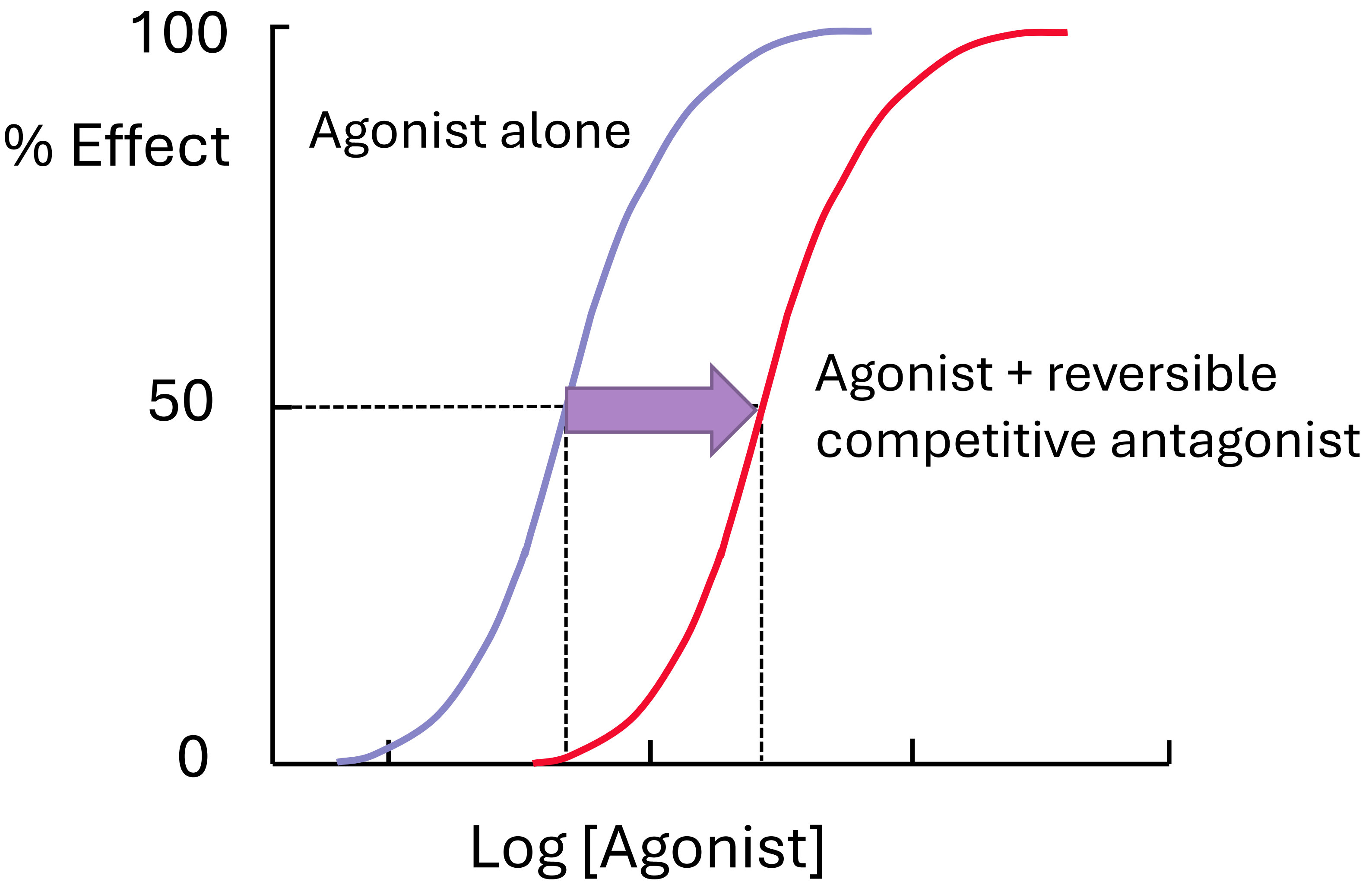
- **So, the agonist concentration-effect curve shifts to the right .**

- High concentrations of agonist can **surmount** the effect of a given concentration of the antagonist: the E_{max} for the agonist remains the same.

Reversible Competitive Antagonism



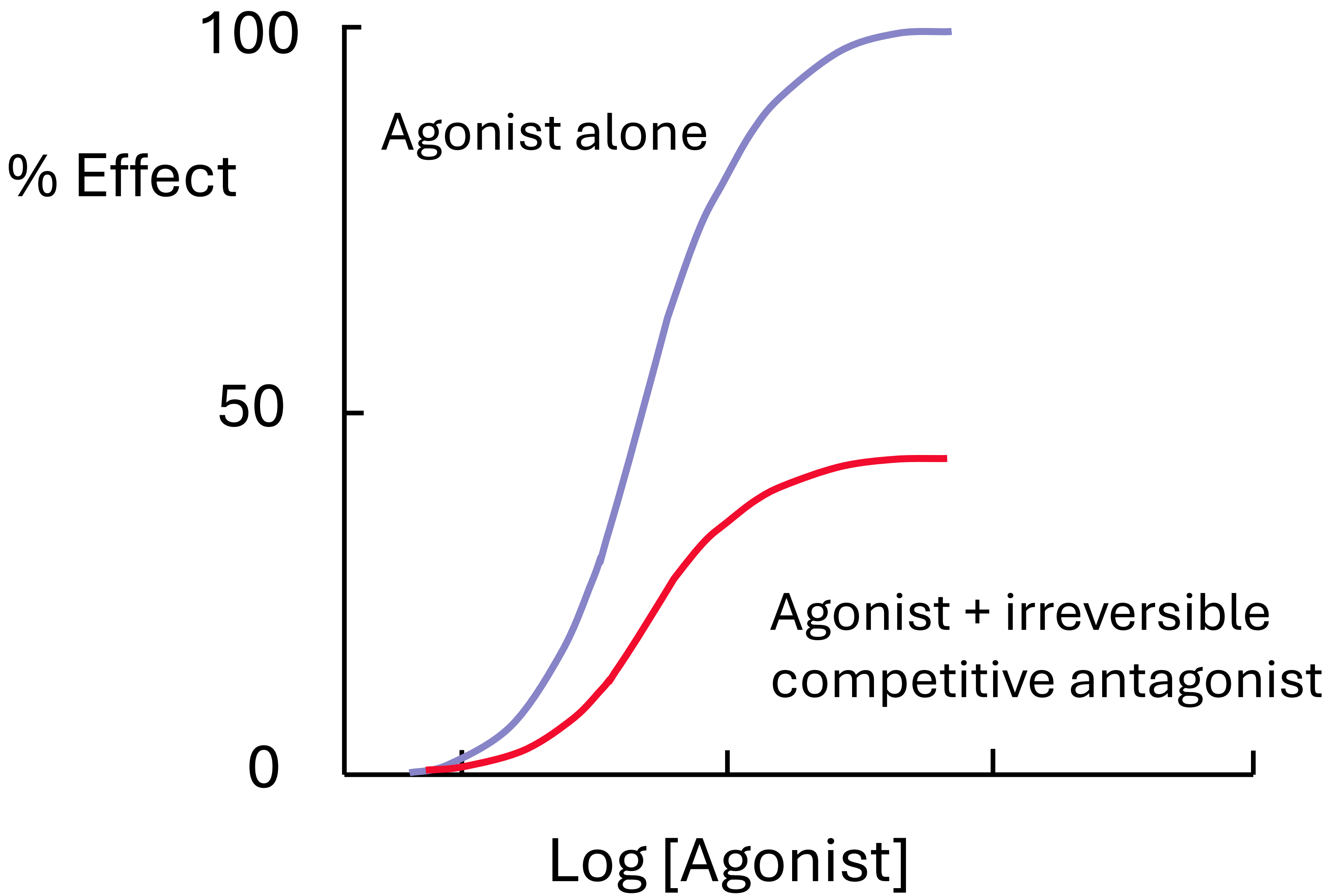
Reversible Competitive Antagonism



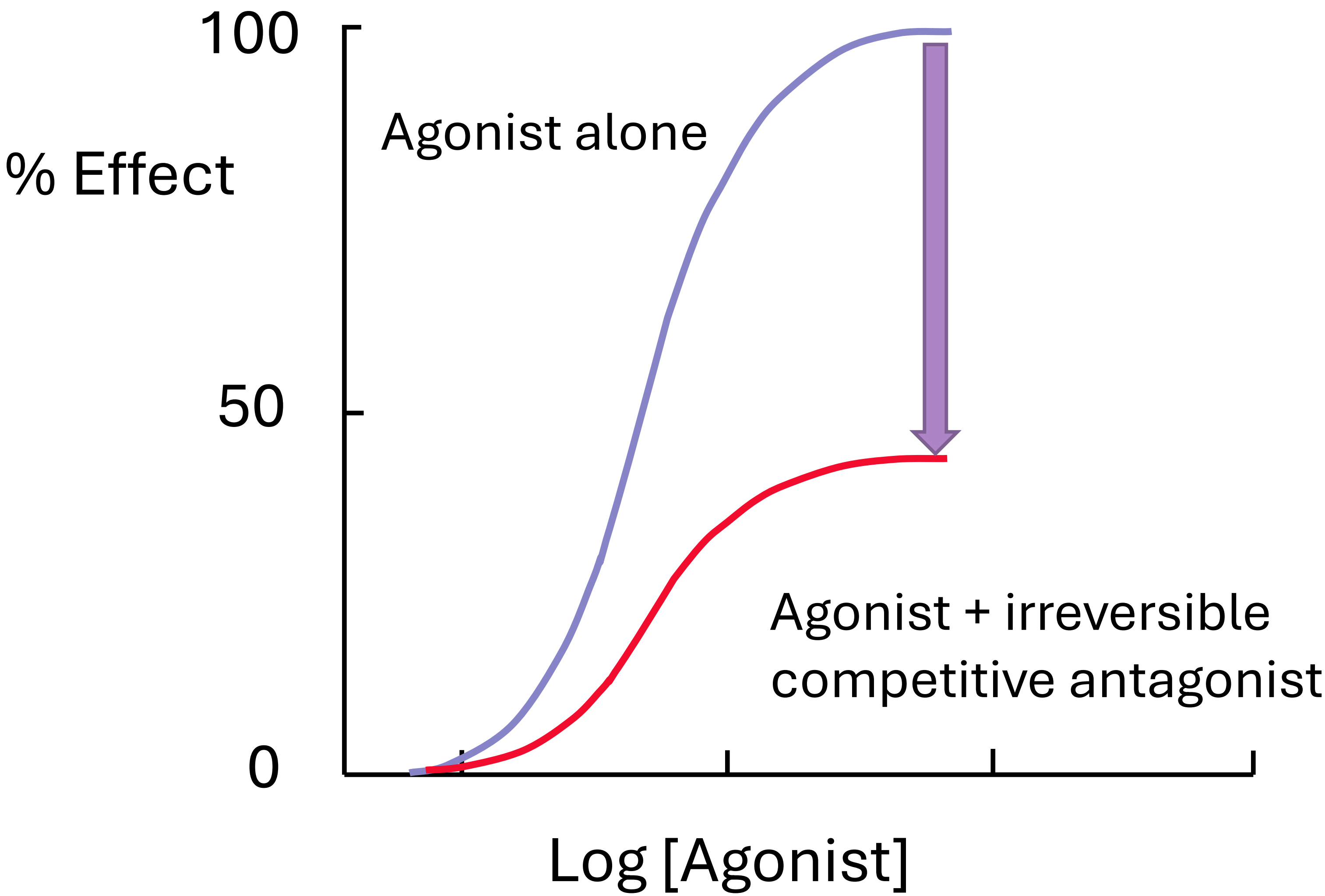
Irreversible Competitive Antagonism

- A receptor bound by an irreversible antagonist cannot respond to the binding of an agonist.
- The E_{max} of the agonist is reduced.
- This antagonism is **insurmountable**.
- ***Some authors refer to this type of antagonism as noncompetitive.***

Irreversible Competitive Antagonism

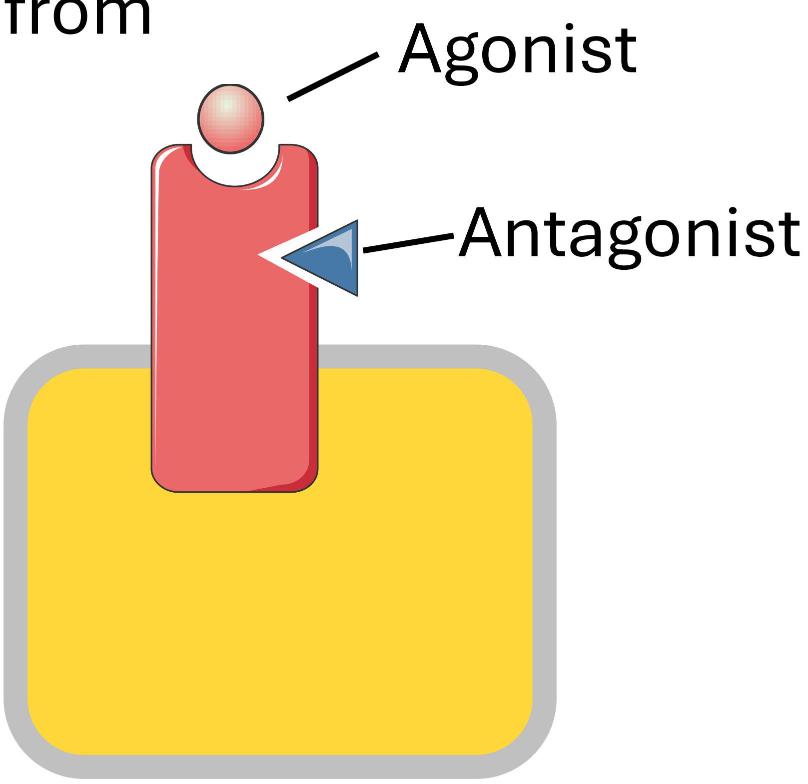


Irreversible Competitive Antagonism

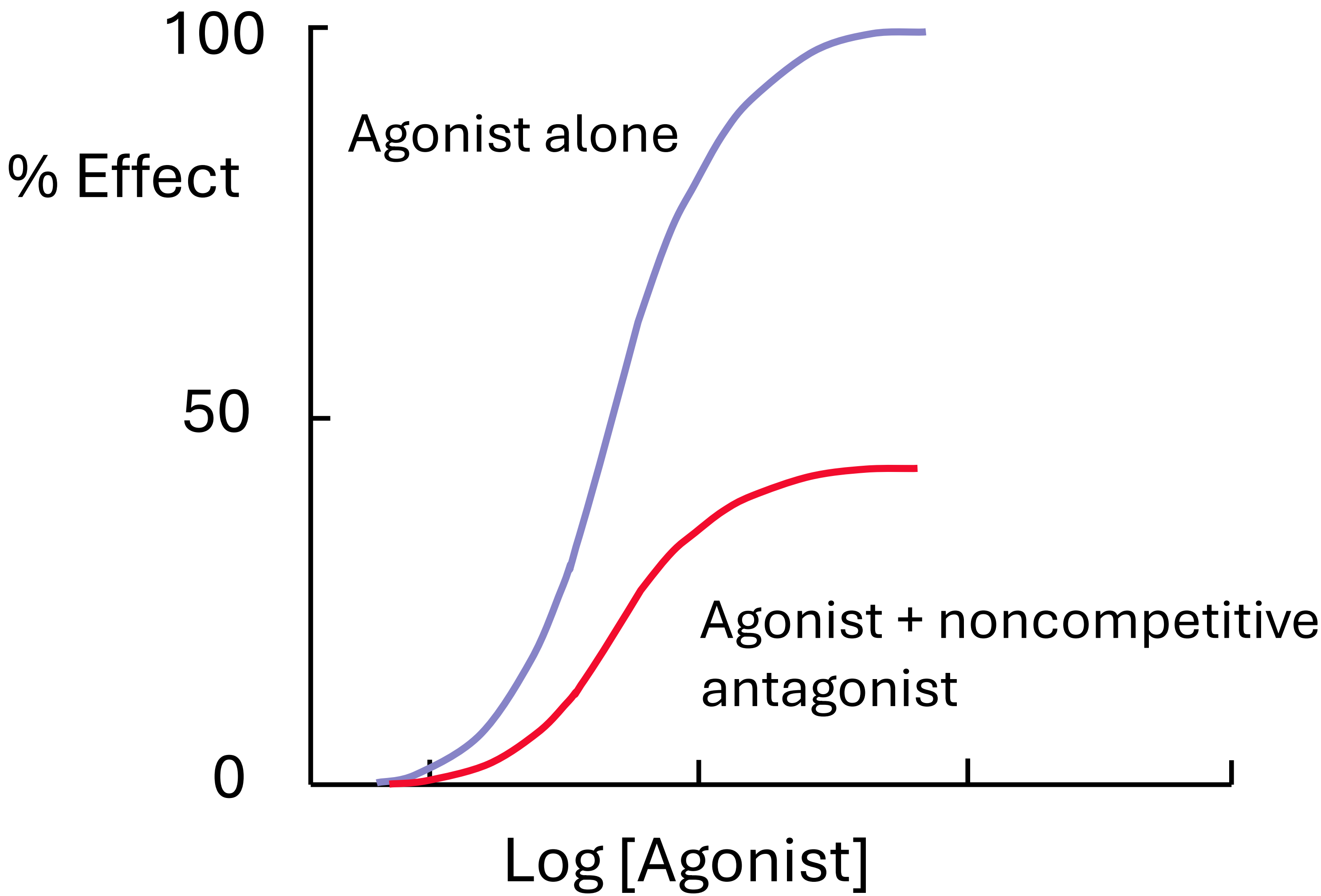


Noncompetitive Antagonism

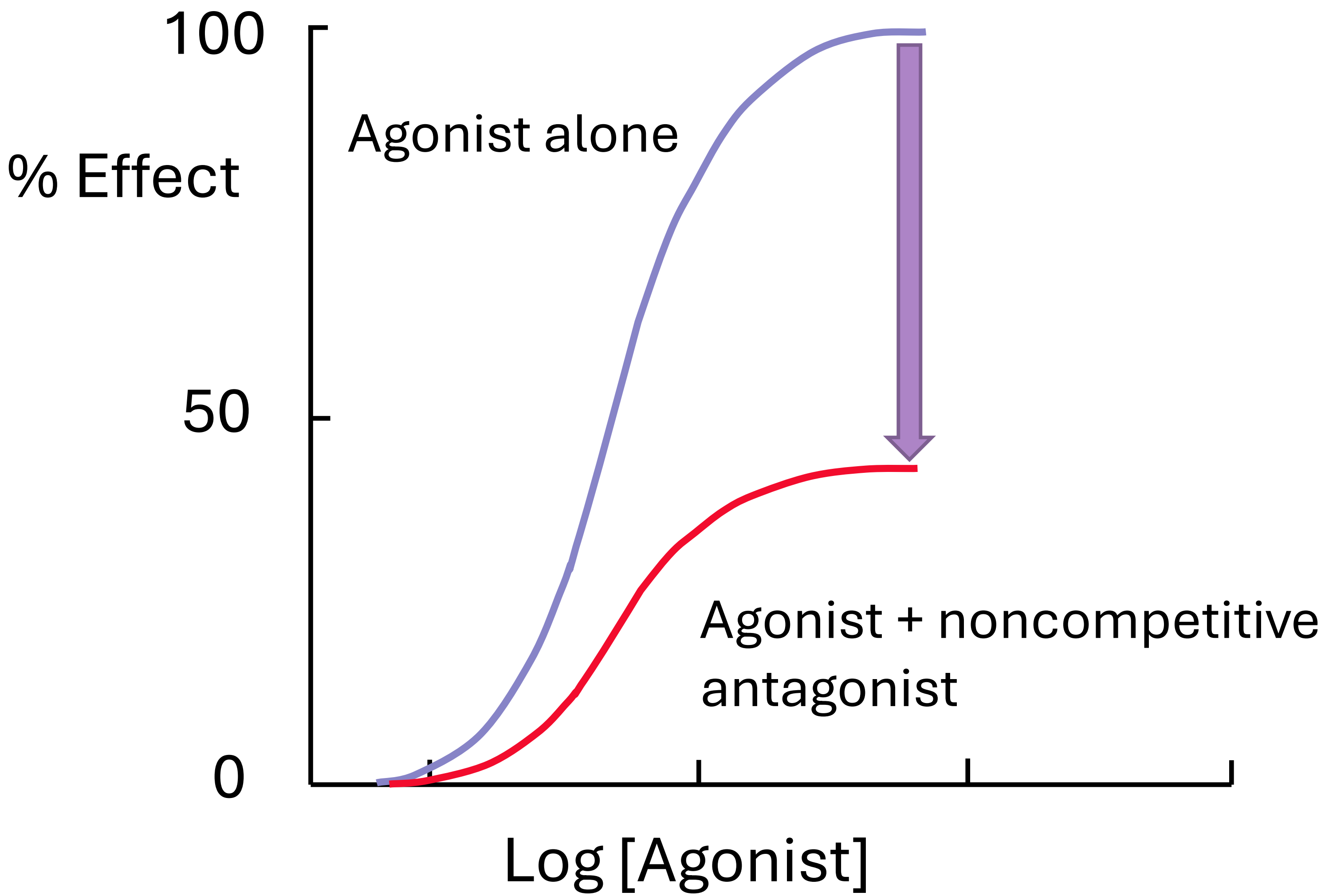
- Also called **allosteric** antagonism.
- Noncompetitive antagonists bind to the receptor at a site different from the agonist binding site.
- This type of antagonism is insurmountable.
- E_{max} is decreased.



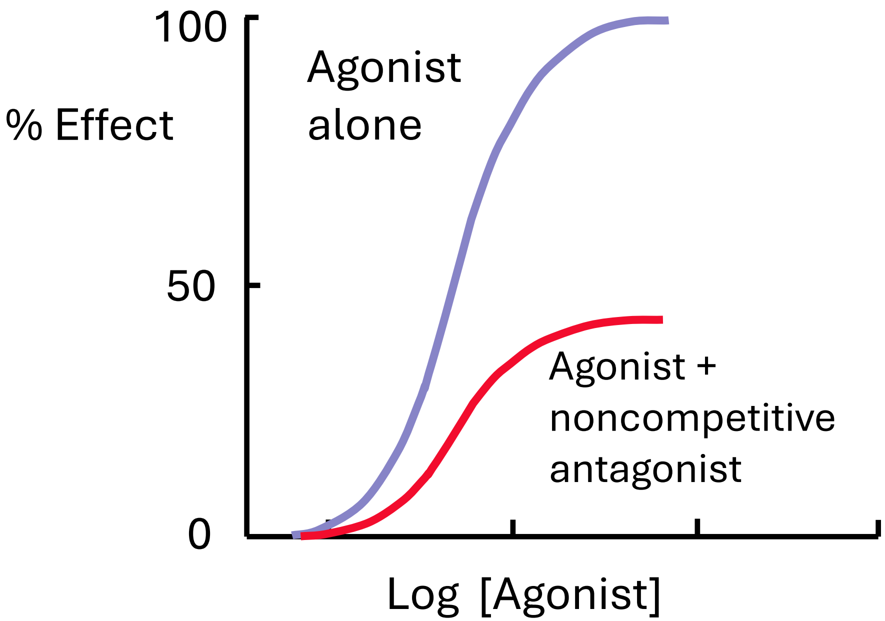
Noncompetitive Antagonism

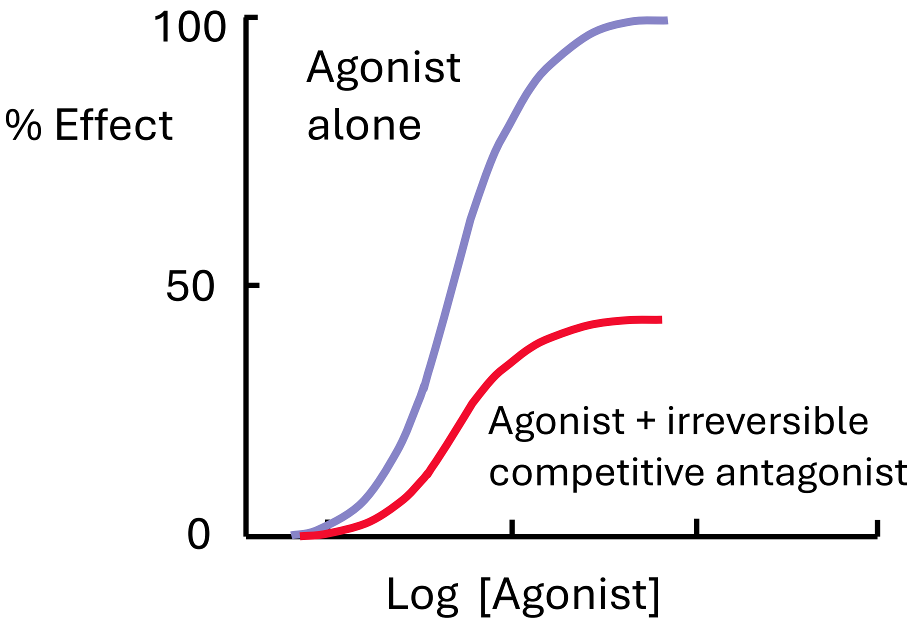


Noncompetitive Antagonism



Irreversible Competitive Antagonism & Noncompetitive Antagonism





NONRECEPTOR ANTAGONISM

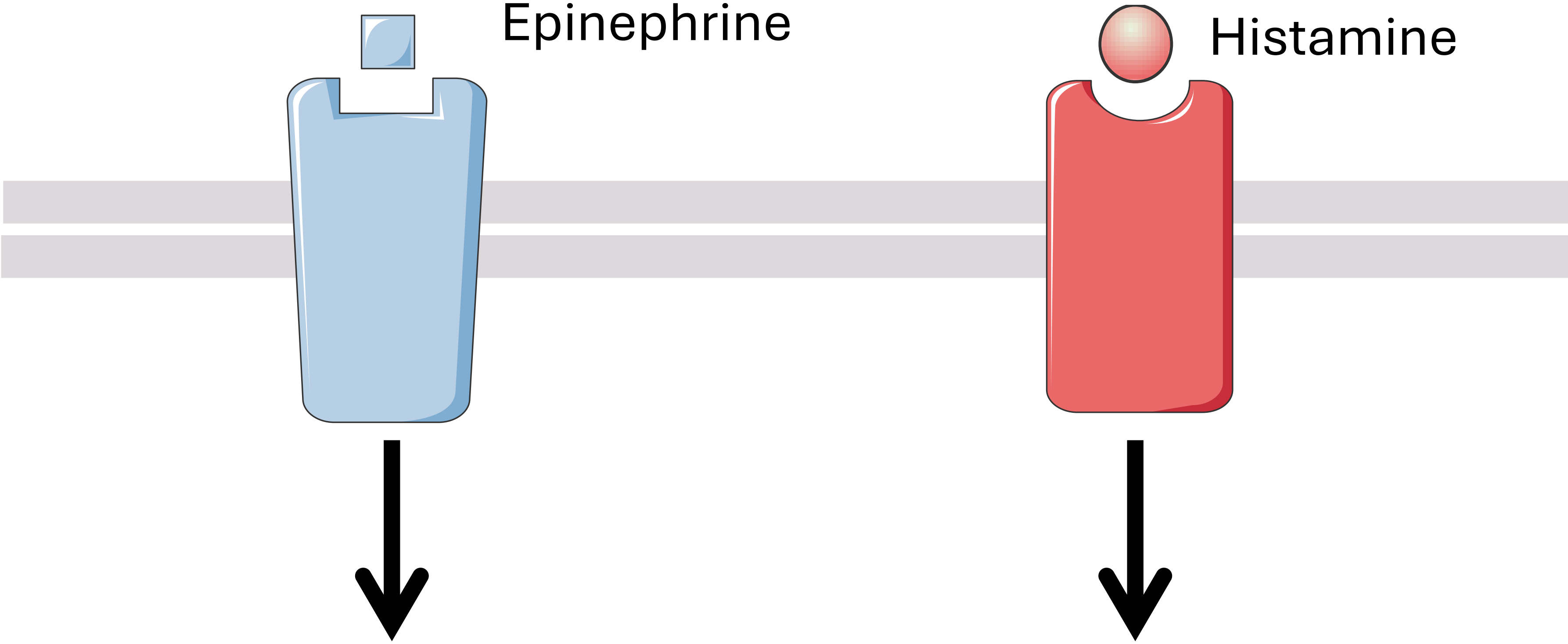
- A nonreceptor antagonist does not bind to the receptor to which the agonist binds.

- Nonreceptor antagonism
  - Physiological antagonism
  - Chemical antagonism

Physiological Antagonism

- One drug opposes another drug, but through **different receptors.**

Physiological Antagonism

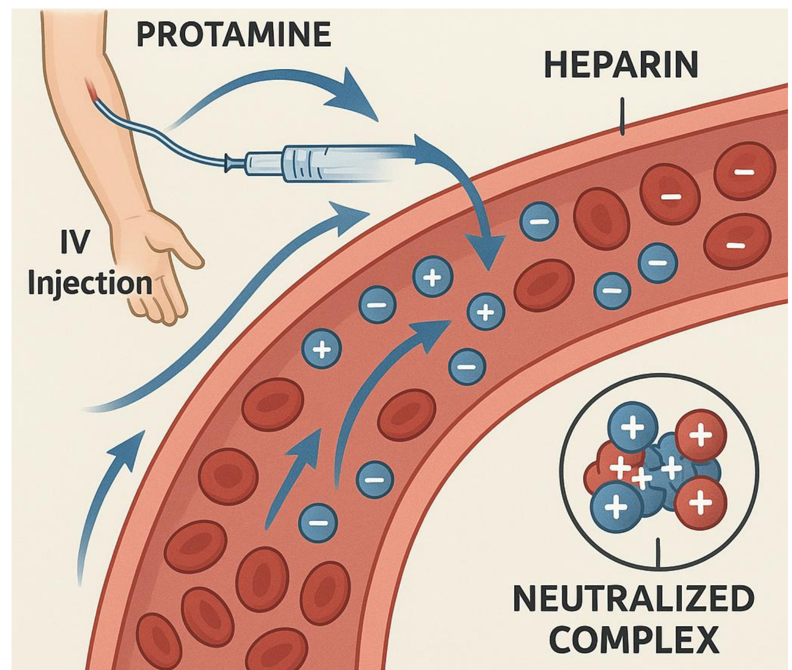


Increase in blood pressure
&
Bronchodilation

Decrease in blood pressure
&
Bronchoconstriction

Chemical Antagonism

- A chemical antagonist reacts chemically with an agonist to form an inactive product.
- For example, **protamine**, a protein which is positively charged, counteracts the effects of **heparin**, an anticoagulant that is negatively charged.



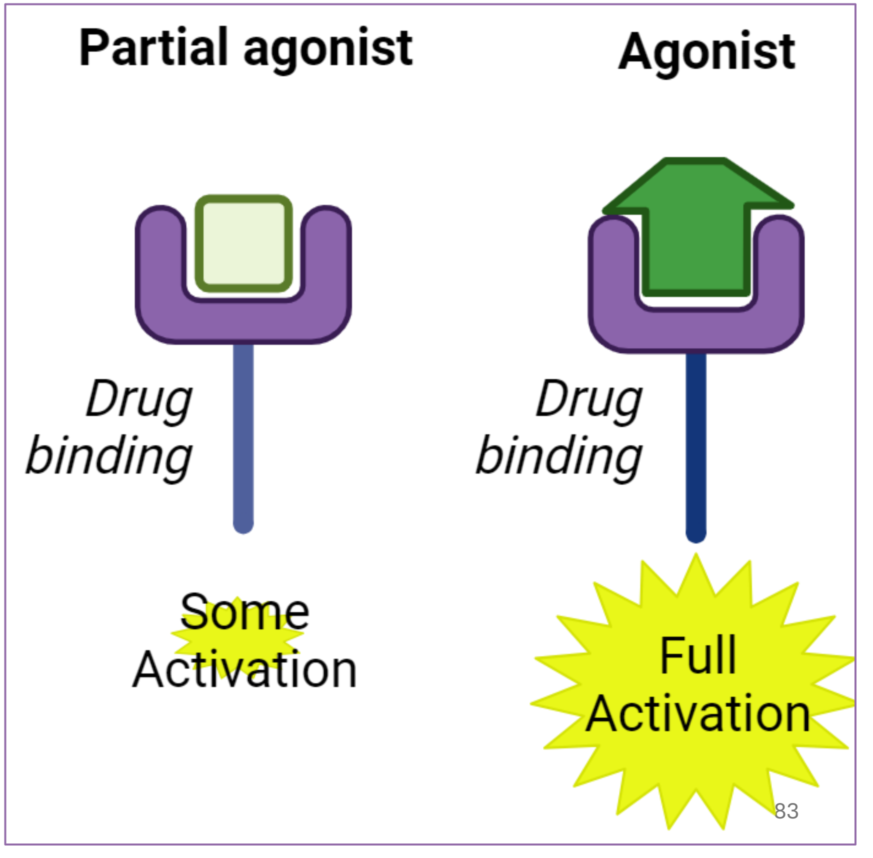
Full & Partial Agonists

- Agonists activate the receptor that they occupy.
- Antagonists cause no activation.
- But the ability of a drug molecule to activate the receptor is a **graded property**.

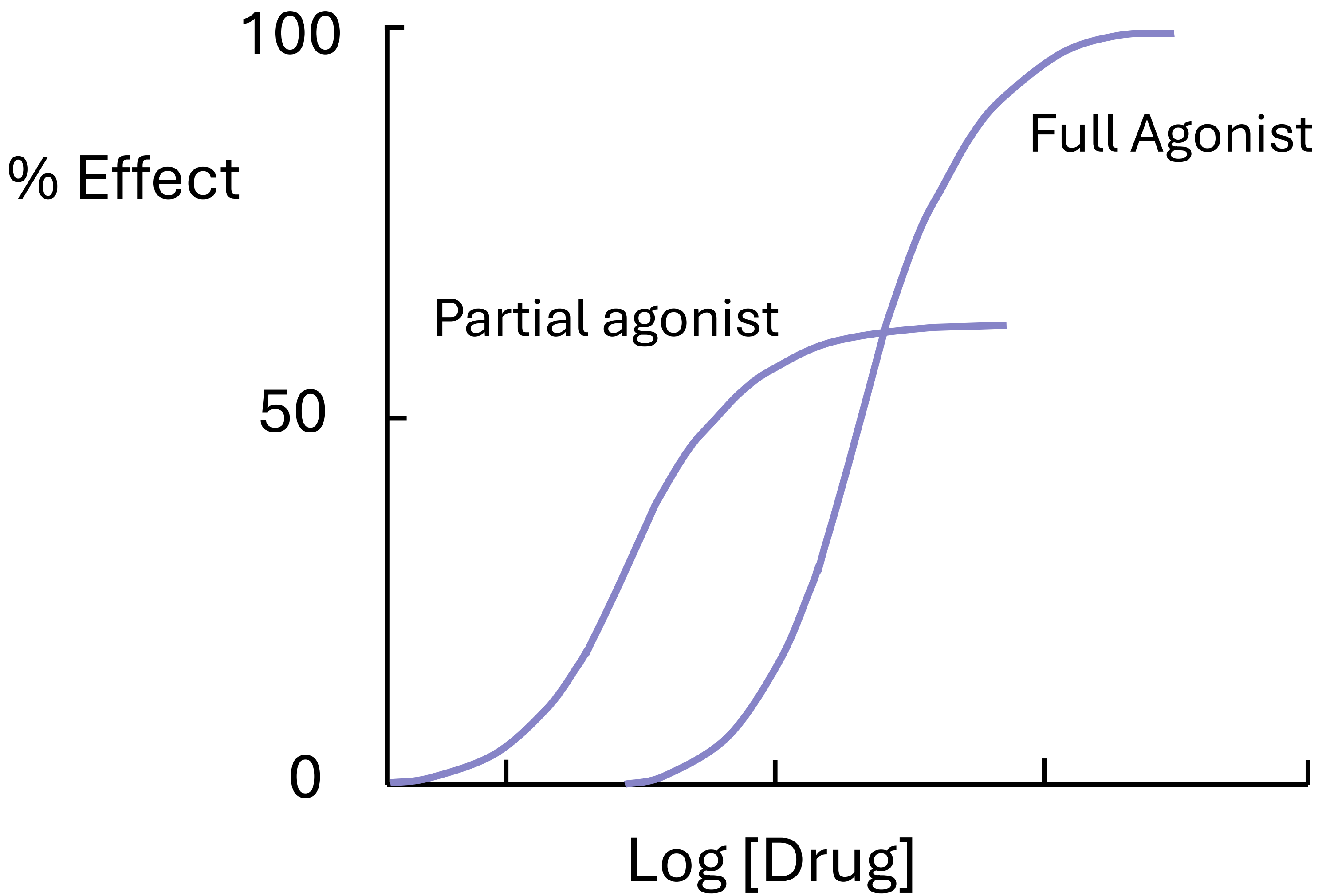
Full & Partial Agonists

- **Full agonists** produce a **maximal response**.
- **Partial agonists** produce a **submaximal response.**

- Partial agonists produce a lower response, at full receptor occupancy, than do full agonists

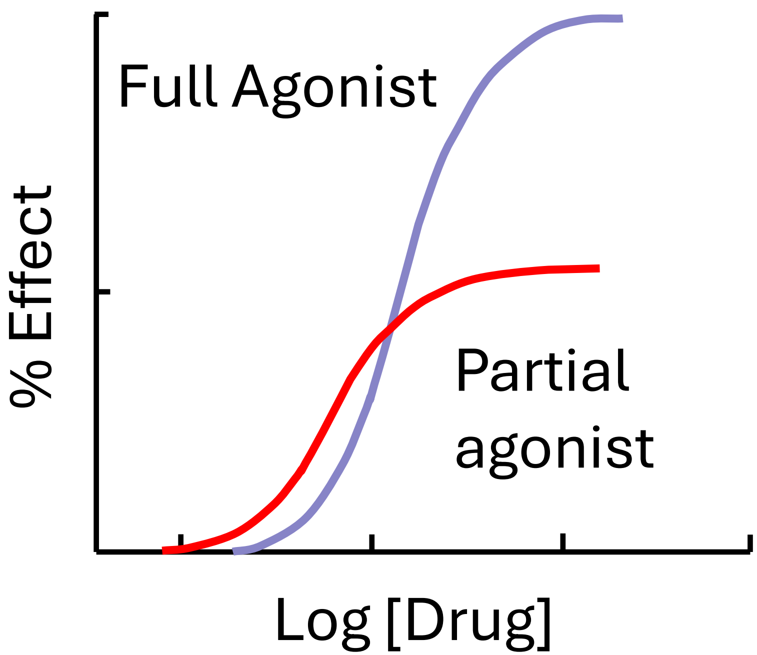


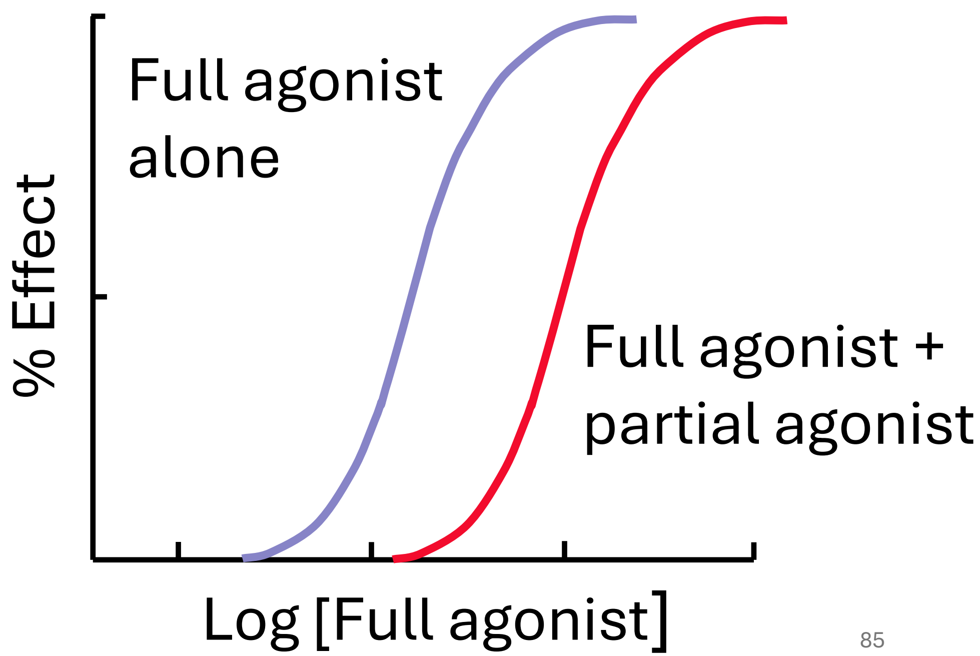
Full & Partial Agonists



Full & Partial Agonists

- A partial agonist can act as a competitive antagonist in the presence of a full agonist, by competing with the full agonist for receptor occupancy, thus reducing the response to the full agonist.



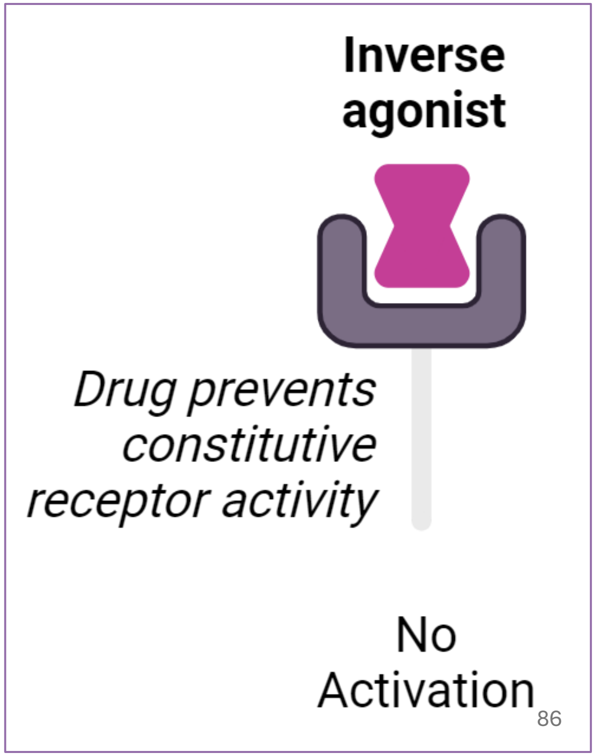


Inverse Agonists

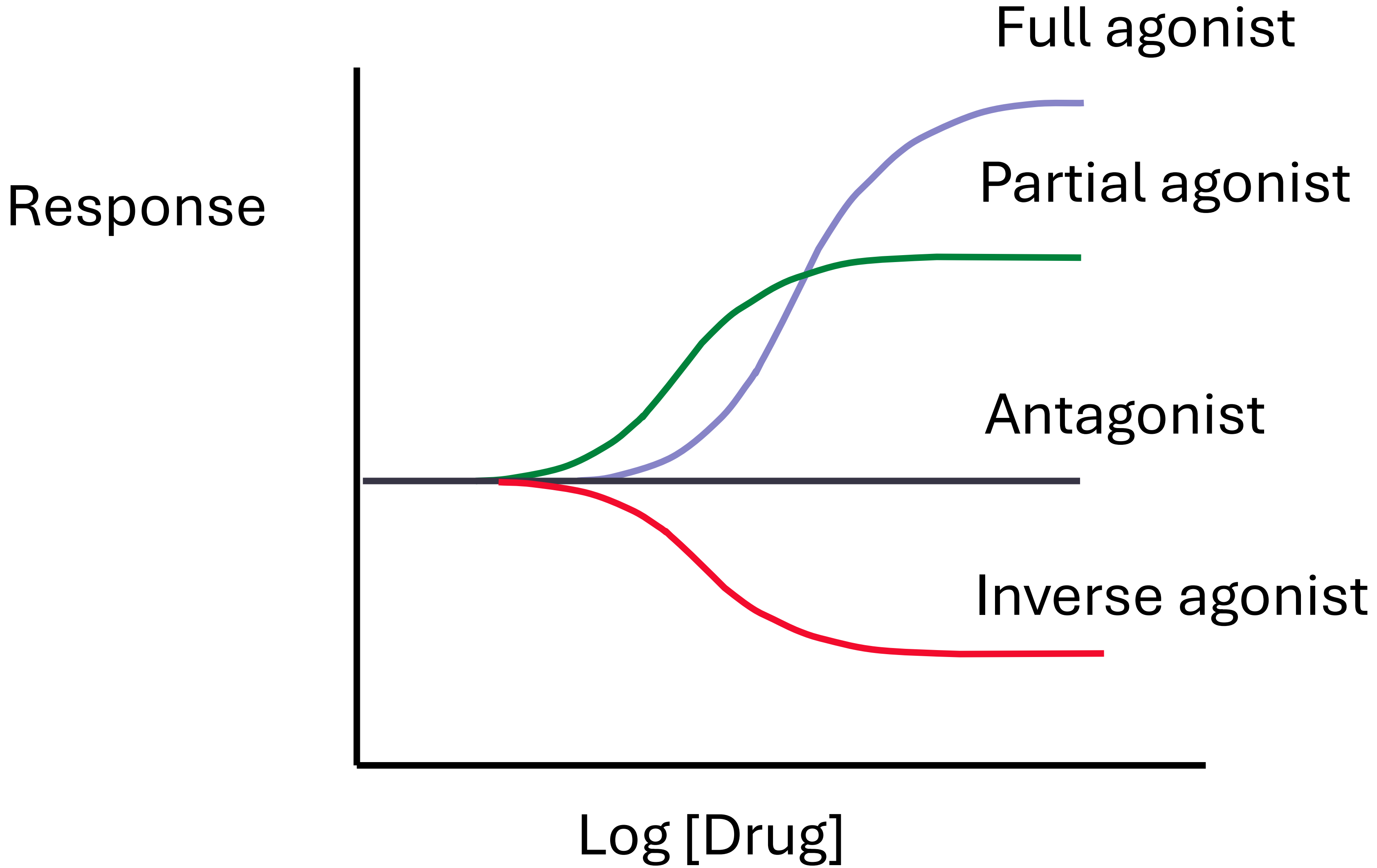
- Many receptor systems exhibit some activity in the absence of an agonist, suggesting that a fraction of the receptors is always in the active state.

- Activity in the absence of an agonist is called constitutive activity.

- **Inverse agonists reverse the constitutive activity of a receptor.**



Effect of various types of ligands on receptor responses



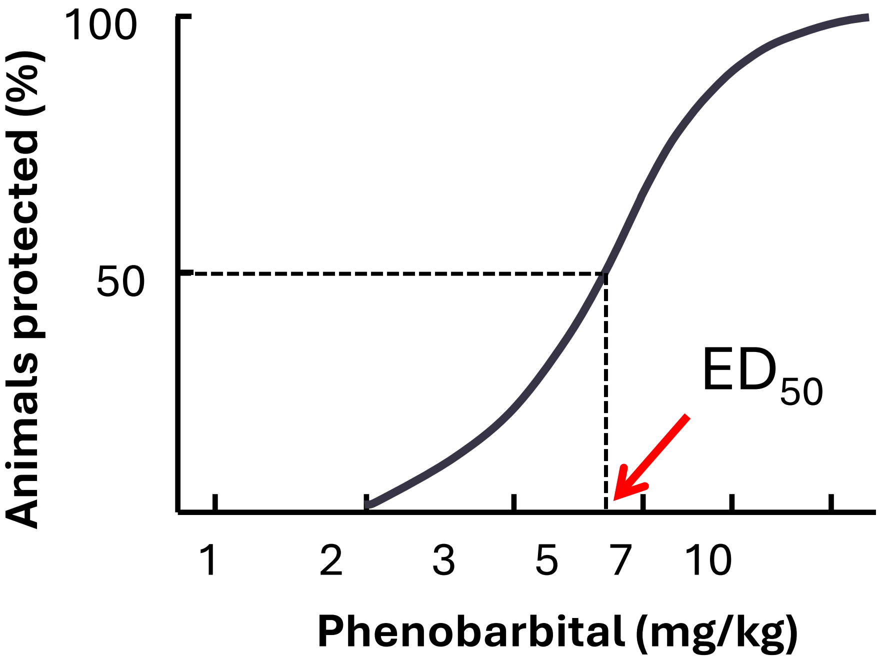
Quantal Dose-effect Curves

- The quantal dose–response relationship plots the **fraction of the population** that responds to a given dose of drug as a function of the drug dose.
- The responses are defined as **either present or not presen**t (i.e., quantal, not graded), such as **prevention of convulsions, arrhythmia, or death.**

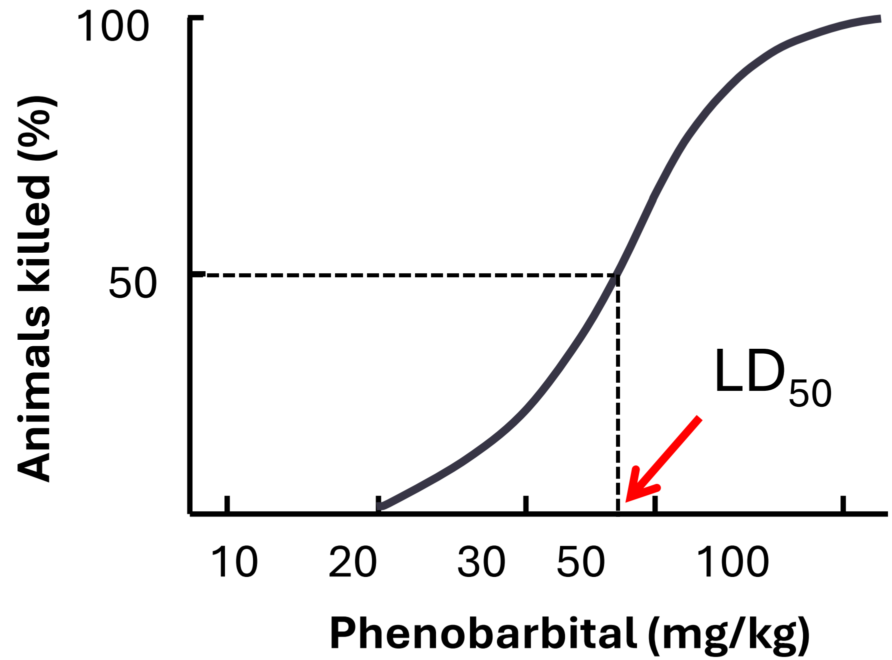
Quantal Dose-effect Curves

- The quantal dose-effect curve is characterized by stating the **median effective dose (ED50):** the dose at which 50% of individuals exhibit the specified quantal effect.
- The dose required to produce a particular toxic effect in 50% of animals is called the **median toxic dose (TD50).**
- The dose required to cause death in 50% of animals is called the **median lethal dose (LD50).**

Quantal Dose-effect Curves



Relationship between dose of phenobarbital and protection of rats against convulsions.



Relationship between dose of phenobarbital and its lethal effects in rats.

Therapeutic Index

- The **therapeutic index is defined as the ratio of the TD_{50} or the LD_{50} to the ED_{50}** for a therapeutically relevant effect.

$$\mathbf{TI} = \frac{\mathbf{TD50}}{\mathbf{ED50}}$$

Or

$$\mathbf{TI} = \frac{\mathbf{LD50}}{\mathbf{ED50}}$$

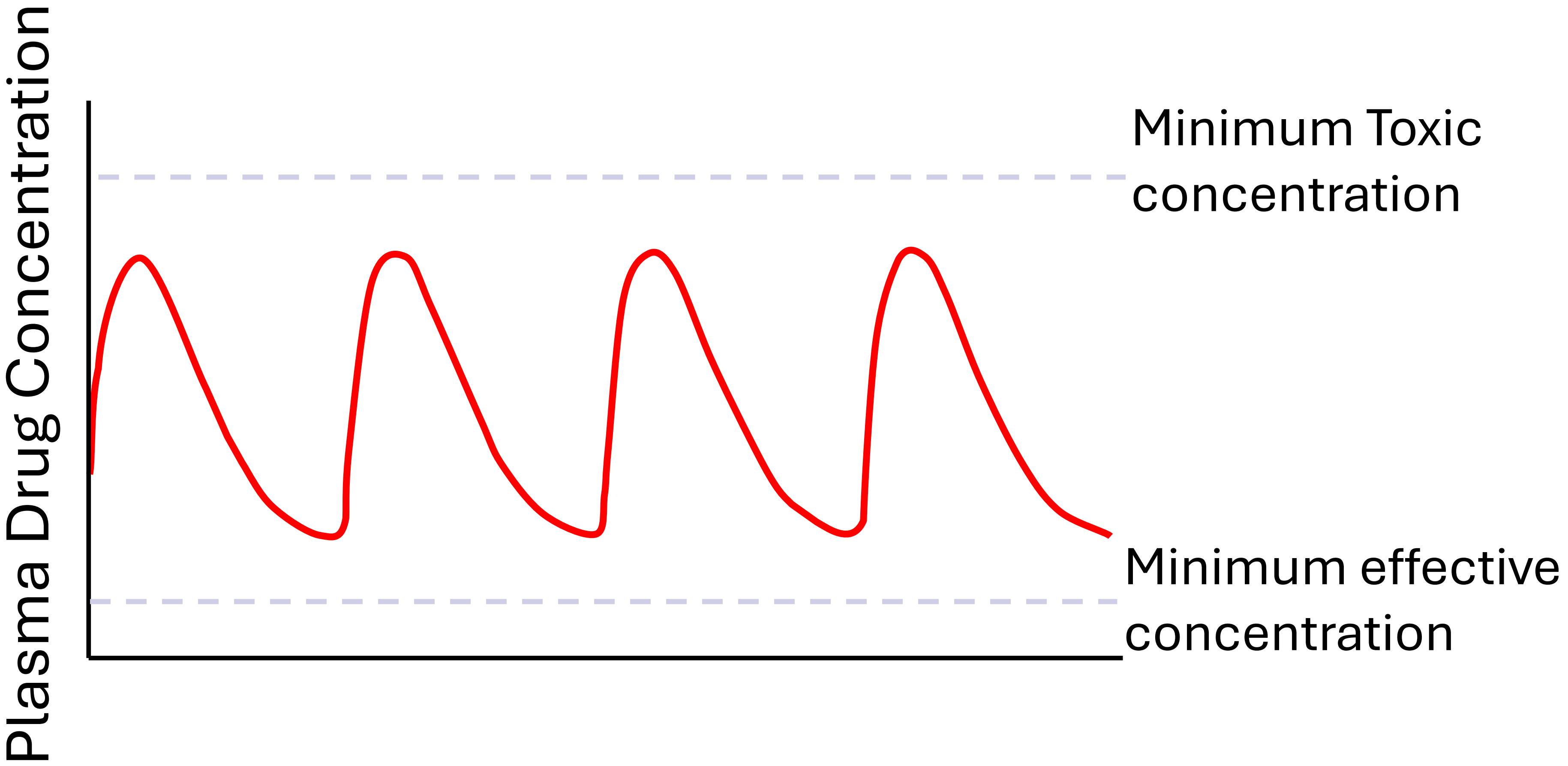
Therapeutic Index

- The therapeutic index of a drug in humans is almost never known with real precision.
- The TI represents an estimate of the safety of the drug.
- **A very safe drug will have a large TI.**

The Therapeutic Window

- The therapeutic window is a more clinically relevant index of safety.
- It is the dosage range between the minimum effective therapeutic concentration and the minimum toxic concentration.

The Therapeutic Window



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